ROBUST LOW-LIGHT OBJECT DETECTION: COMBINING OFFLINE ILLUMINATION COMPENSATION, VISION LANGUAGE KNOWLEDGE DISTILLATION AND REAL-TIME DETECTION TRANSFORMERS

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THESIS SUBMITTED IN FULFILMENT OF THE REQUIREMENTS FOR THE DEGREE OF MASTERS OF ARTIFICIAL INTELLIGENCE

FACULTY OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY
UNIVERSITI MALAYA
KUALA LUMPUR

2026

ABSTRACT

Object detection models trained under standard daytime conditions suffer from significant accuracy degradations when deployed in low-light environments, due to reduced contrast, sensor noise, and illumination distortions, thus presenting safety-critical challenges for applications such as nighttime autonomous driving and drone surveillance. Existing research in this field falls into three categories: joint enhancement-detection, real-time low-light Detection Transformers (DETRs), and knowledge distillation methods using Vision Language Models (VLMs). This research proposes a decoupled, three-stage pipeline that combines offline illumination compensation with VLM-based knowledge distillation to train a real-time DETR for low-light object detection. To isolate the individual and combined contributions of spatial enhancement and semantic distillation, several experimental conditions are compared: a no-intervention baseline trained on raw images with human labels, a spatial-enhancement-only baseline trained on enhanced images with human labels, and the proposed condition trained on enhanced images with VLM pseudo-labels, as well as hybrid human and pseudo-labels. Full human supervision on raw images performed best at 0.893 mAP@0.5 and 0.634 mAP@[.5:.95], with enhanced human labels marginally behind at 0.883 mAP@0.5 and 0.621 mAP@[.5:.95]. The hybrid trained models followed, at 0.827 mAP@0.5 and 0.479 mAP@[.5:.95] for a 25% of human labels mixed into the pseudo label set, and 0.799 mAP@0.5 and 0.449 mAP@[.5:.95] for 10% of human labels mixed. Pseudo-labels alone placed lowest among the trained conditions, at 0.758 mAP@0.5 and 0.420 mAP@[.5:.95], for the 0.55 confidence threshold. Enhancement produced no measurable benefit in this experiment, where Zero-DCE provided no significant contribution in improving mAP.

Keywords: low-light object detection, knowledge distillation, vision-language models, detection transformers, pseudo-labelling

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LIST OF SYMBOLS AND ABBREVIATIONS

AIFI	:	Attention-based Intra-scale Feature Interaction
AP	:	Average Precision
CCFF	:	Cross-scale Feature Fusion
CDRC	:	Cross-Domain Representation Compensating module
CLAHE	:	Contrast Limited Adaptive Histogram Equalization
CNN(s)	:	Convolutional Neural Network(s)
COCO	:	Common Objects in Context (dataset)
ConCal	:	Pseudo Label Confirmation Calibrating module
DETR(s)	:	DEtection TRansformer(s)
DeT	:	Debiased Teacher
DFFAM	:	Dual-branch Feature Fusion Attention Module
DINOv2	:	Self-supervised vision transformer backbone, version 2
DNDT	:	Day-to-Night Domain Transforming module
EMA	:	Exponential Moving Average
ExDark	:	Exclusively Dark (image dataset)
FFT	:	Fast Fourier Transform
FN	:	False Negative
FP	:	False Positive
FPA	:	Frequency and Physics-Aware module
FPS	:	Frames Per Second
GFLOPs	:	Giga Floating-point Operations
GT	:	Ground Truth

IoU	:	Intersection over Union
KDFA	:	Knowledge distillation with geometry-consistent Feature Alignment
LDETR	:	Low-light DEtection TRansformer
LLAM	:	Low-Light Adaptive Module
MACs	:	Multiply-Accumulate operations
mAP	:	mean Average Precision
MM-GroundingDINO	:	Multi-Modal Grounding DINO
NAS	:	Neural Architecture Search
NF-DETR	:	Night-Frequency Detection Transformer
NMS	:	Non-Maximum Suppression
OWLv2	:	Open-World Localization, version 2
RF-DETR	:	Roboflow DEtection TRansformer
RT-DETR	:	Real-Time DEtection TRansformer
SCI	:	Self-Calibrated Illumination
SOTA	:	State-Of-The-Art
TP	:	True Positive
VLM(s)	:	Vision-Language Model(s)
VLMOE	:	VLM-Based Orthogonality Enhancement
WCAF	:	Window Cross-Attention Fusion
YOLO	:	You Only Look Once
Zero-DCE(++)	:	Zero-Reference Deep Curve Estimation (++)

CHAPTER 1: INTRODUCTION

1.1 Introduction

Object detection models trained under standard daytime conditions experience accuracy degradation when deployed in low-light environments, due to reduced contrast, photonic sensor noise, and illumination distortions that negatively affect semantic and spatial features. This is especially critical for industry applications such as nighttime autonomous driving and drone surveillance, where accuracy and latency are critical to ensure safety.

One common approach to this problem has been to implement image enhancement directly into the detection pipeline, either as a pre-processing stage or as a module jointly trained with the detector. While such approaches can improve visibility under low-light conditions, the enhancement module is typically coupled tightly to a specific detector backbone, making the resulting pipeline brittle, where changes to the detector architecture requires retraining the entire system from scratch. This limits the reusability of enhancements across different detector designs.

Another line of development focuses on focuses on the detection architecture itself. YOLO-family models rely on Non-Maximum Suppression (NMS) as a post-processing step, which introduces latency and many-to-one bounding box assignment strategies. Real-time detection transformers (DETRs) eliminate this post-processing step entirely through end-to-end, one-to-one bipartite matching, enabling a simpler prediction architecture. The self-attention mechanism of these transformers also captures global context and long-range dependencies, motivating their adoption in low-light detection where degraded local cues benefit from broader global context.

A third strategy is knowledge distillation, where a larger, more capable teacher model's representations are transferred to a simpler student detector model, shifting the

out the complexity of the detector architecture to the knowledge distillation process at training. Vision Language Models (VLMs) offer a way to conduct robust distillation. Pre-trained on internet scale text and image data, these models possess zero-short semantic grounding that make them potential candidates as detection teachers that generate pseudo-labels for distillation. Additionally, the one-to-one pseudo-labels are also naturally compatible with NMS-free detection transformers, making it a suitable candidate to couple with DETRs to further drive detection capabilities.

This research proposes an approach that combines the approaches above, starting from a decoupled pipeline that incorporates a physics-based illumination compensation module, and a VLM-based knowledge distillation architecture to generate pseudo-labels to train a DETR model for low-light object detection. Using an empirical comparison approach between a no-intervention baseline, a illumination enhancement only baseline, and a VLM distillation condition under similar conditions, the study investigates the individual contributions of illumination compensation and semantic distillation when deployed in a single detection pipeline.

1.2 Research Problems

1. How does the choice of VLM teacher and the physics-based illumination method affect the VLM-generated pseudo-labels for low-light object detection? Specifically, do VLM teachers produce differently calibrated confidence scores on raw versus spatially enhanced images, and how does this difference affect the quality of pseudo labels?
2. When decoupled from the detector architecture, does offline spatial enhancements introduce synthetic noise artefacts that degrade the quality of VLM teacher pseudo labels? Can VLM semantic regularisation ability mitigate this degradation relative to a spatial-enhancement only baseline?

3. How does the VLM knowledge distilled real time DETR compare to YOLO-family detectors and internally modified transformer architectures on low light detection benchmarks, under a standardised evaluation protocol?

1.3 Research Objectives

1. To empirically select and evaluate a suitable VLM annotation teacher from open-source models by comparing VLM performance on low light benchmark datasets. Perform evaluation by measuring per-class confidence distribution, precision and IoU against ground truth.
2. To evaluate the individual and combined contributions of spatial enhancement and knowledge distillation to low-light detection accuracy by comparing real time DETR performance trained using raw human labels, physics enhanced human labels, and enhanced VLM labels.
3. To benchmark the trained DETR results against published results on the evaluated dataset, based on mAP, parameter count, GLFOPs and latency.

CHAPTER 2: LITERATURE REVIEW

2.1 Enhancement Strategies for Low-Light Detection

2.1.1 From Enhance-Then-Detect to Joint Learning Frameworks

Advances in low-light computer vision research has shifted from simple "enhance-then-detect" pipelines, where techniques such as histogram equalisation are deployed on the image in isolation before fed to the detection model. These methods struggle to learn task-specific features and can amplify noise or artifacts that harm detection performance. Additionally, such serial pipelines can incur high computational costs, presenting challenges for real-time edge deployment.

Instead, recent research emphasises end-to-end joint learning frameworks, where enhancement and detection modules are trained simultaneously. For instance, Dec-YOLO by Li et al. (2026) combines a DeNet pre-backbone enhancement module that is optimized simultaneously with the object detection module. Based on Retinex theory, this module treats an image as a product of illumination and reflection, allowing the model to learn and reconstruct the target features through a Dynamic Adaptive Contract Reconstruction Network (DACR Net), as opposed to uniformly increasing the scene brightness.

Liu et al. (2025) proposed a parallel joint encoding approach to tackle low-light detection for drone scenarios. In this setup, two primary subnetworks: a YOLOv5 detection baseline, and an image enhancement module using Zero-DCE++ (Zero-reference Deep Curve Estimation ++) are integrated through shared gradients and joint optimisation. Zero-DCE++ utilises unsupervised learning to perform adaptive luminance correction to adjust the brightness distribution and luminance of an image.

2.1.2 Integration of Specialised Modules

Alongside joint learning frameworks, a parallel theme in state-of-the-art research is the integration of specialised enhancement modules within detection architectures. Zhao et al. (2025) proposed the DAMFCN network for low-light image enhancement. The network consists of a Low-Light Adaptive Module (LLAM), which extracts rich semantic information and suppresses noise artefacts introduced during image brightening, alongside a Dual-branch Feature Fusion Attention Module (DFFAM) that mitigates blur and occlusion effects. DAMFCN is paired with a DarkYOLOv8 detector, achieving an extremely lightweight profile of 2.9M parameters and 6.7 GFLOPs.

2.1.3 Limitation of Enhancement Modules: Coupling and Pipeline Brittleness

Such integrated pipelines such as DeC-YOLO (Li et al, 2026) allow detection gradients to guide enhancement decisions, where the gradient penalises the model if it brightens a region that does not help detection. However, this introduces a critical limitation, where the enhancement module becomes inseparable from the detection model it is attached to. As a result, the DACR-Net in DeC-YOLO cannot be transplanted to a different detector, making the pipeline brittle. Every detector change or upgrade requires retraining of the entire pipeline from scratch.

2.2 Architectural Advancements towards Real-Time Transformers

From an architectural development perspective, research has undergone a transition from object detection using Convolutional Neural Networks (CNNs) to specialised transformer-based Detection Transformers (DETRs), which excel in capturing global context and long-range dependencies.

2.2.1 Transformer-Based Architectures for Low-Light Detection

Tiwari et al. (2024) proposed the Low-light DEtection TransformeR (LDETR), an end-to-end transformer architecture specifically designed for object detection in low-light and

adverse weather conditions. Unlike traditional DETRs, this model utilises two encoder-decoder pairs: the first is dedicated to discovering the intrinsic visual structure of the scene and efficiently discriminating between foreground objects and noisy backgrounds, while the second performs the object detection task itself.

Li and Shen (2025) introduced NF-DETR (Night-Frequency Detection Transformer), which processes images through two parallel streams via its Frequency and Physics-Aware (FPA) module. The first stream extracts the reflectance component of an object using Retinex theory, similar to DeC-YOLO. The second stream applies Fast Fourier Transform (FFT) to convert spatial features into the frequency domain, focusing on high-frequency cues that represent edges and contours. Features from both streams are fused using Window Cross-Attention Fusion (WCAF).

2.2.2 Evolution towards Real Time Detection Transformers

Real-time DETRs represent a shift in object detection, from CNN-based architectures such as YOLO towards end-to-end transformers that retain high accuracy while providing the inference speeds required for real-world applications.

Zhao et al. (2024) presented RT-DETR, which replaces the vanilla transformer encoder with a hybrid encoder that splits feature processing into Attention-based Intra-scale Feature Interaction (AIFI), performing self-attention only on high-level features, and Cross-scale Feature Fusion (CCFF) for multi-scale fusion. The defining feature of RT-DETR is the elimination of Non-Maximum Suppression (NMS), a post-processing step required by YOLO models to remove overlapping bounding boxes. NMS introduces computational bottlenecks and instability due to sensitivity to hand-crafted hyperparameters such as confidence and IoU thresholds.

Building on RT-DETR, RT-DETRv2 (Zhao et al., 2024) introduced multi-scale sampling flexibility, allowing a distinct number of sampling points for features at different scales, and improved hardware compatibility through a discrete sampling operator.

RT-DETRv3 (Wang et al., 2025) addresses the sparse supervision signal generated by the one-to-one set matching strategy, in which each ground truth object is matched to only a single predicted query during training. This leads to inadequate training of both the encoder and decoder. Improvements include a one-to-many label assignment head operating in parallel with the transformer branch during training, and random masking to perturb the self-attention mechanism.

RF-DETR (Robinson et al., 2026) is a lightweight specialised object detection transformer, reported as the first real-time detector to exceed 60% Average Precision on the COCO benchmark dataset. Its primary enhancement is the integration of weight-sharing Neural Architecture Search (NAS) with the DINOv2 backbone.

2.2.3 Overcoming YOLO NMS Benchmarking Asymmetry

A systemic benchmarking asymmetry in YOLO-based detectors, as reported by Robinson et al. (2025), is noteworthy. YOLO detection accuracy is reported with NMS applied, while inference latency is reported without. This means the efficiency advantage of the YOLO family in low-light applications may have been overstated. Furthermore, NMS from YOLO-family models introduces structural incompatibilities when paired with VLMs that produce one-to-one pseudo labels. NF-DETR (Li and Shen, 2025) partially resolves this by embedding a custom FFT module into the transformer backbone, at the cost of architectural simplicity and increased GFLOPs.

2.3 Knowledge Distillation and Domain Adaptation

A complementary approach to architectural improvement is knowledge distillation using Vision Language Models (VLMs), where a more powerful model's robust representations are transferred to a student detector. This is possible because VLMs are pre-trained on billions of diverse image-text pairs, granting them zero-shot semantic intelligence that generalises beyond the lighting conditions present in standard training data. Three representative approaches illustrate a progression from supervised paired data, through unsupervised domain adaptation, to self-training.

2.3.1 Supervised Paired Distillation

Shi et al. (2025) implemented knowledge distillation with geometry-consistent feature alignment (KDFA) for low-light apple detection, aiming to transfer knowledge from a daylight teacher to a low-light student. KDFA leverages synthetically generated pixel-aligned image pairs — identical scenes rendered in both low-light and daylight conditions. The teacher, with its weights frozen, sees only the daylight images, while the student is trained on the corresponding low-light pairs. The teacher's feature vectors for each detected region become positive targets towards which the student's representations are pulled via InfoNCE contrastive loss. KDFA achieves 51.3% mAP@[0.50:0.95] on its low-light apple benchmark, improving performance by 12.7 percentage points over a YOLOv7 baseline.

2.3.2 Unsupervised Domain Adaptation with VLM Regularisation

He et al. (2025) implemented unsupervised domain adaptation, training on a labelled daytime dataset and an unlabelled nighttime dataset. A key feature is the VLM-Based Orthogonality Enhancement (VLMOE) module, which uses a frozen VLM encoder to generate visual embeddings for each category. These embeddings are used to force the detector's feature representations to be orthogonal to other categories in the VLM

embedding space, enforcing separability and improving decision boundaries without requiring nighttime labels. He et al. achieved up to 5.5 mAP improvement over state-of-the-art nighttime adaptation methods on BDD100k, FLIR, and SHIFT benchmarks, with no additional inference costs.

2.3.3 Self-Training with Debiased Pseudo Labels

Cui et al. (2025) proposed a Debiased Teacher framework for domain adaptive day-to-night detection, employing a self-training setup where both teacher and student models are Faster R-CNN models with a ResNet-50 backbone. The framework introduces a Day-to-Night Domain Transforming module (DNDT) that generates synthetic night-like images from labelled daytime data, a Cross-Domain Representation Compensating module (CDRC) that performs bidirectional mixing between nighttime and night-like images combined with Inverse Class Frequency Balancing to address class imbalance, and a Pseudo Label Confirmation Calibrating module (ConCal) that replaces fixed confidence thresholds with dynamic per-class thresholds and recovers potential true positives by jointly distilling predictions from both teacher and student when their outputs agree. Cui et al. achieved 50.9% mAP on BDD100k, 52.8% on SHIFT, and 46.7% on the more challenging TDND dataset.

2.3.4 The Teacher Knowledge Ceiling

A review of the knowledge distillation literature reveals that existing teacher models are bounded by the same knowledge ceiling as the student. He et al. (2025) and Cui et al. (2025) use visual-only detectors trained on labelled daytime images as teacher models; these models are therefore extrapolating beyond their training distribution when generating supervision signals for nighttime images.

He et al. (2025) partially addresses this by incorporating a frozen CLIP encoder, but its role is restricted to category-level orthogonality enforcement and does not provide

bounding box predictions or instance-level supervision. Shi et al. (2025) offers a more structurally rigorous teacher-student approach but requires pixel-aligned paired daylight and low-light images of the same scene. Across all approaches, confidence calibration remains unexamined. No validation was performed to confirm that teacher confidence scores reliably correspond to prediction quality under enhanced or degraded inputs.

2.4 Summary of State-of-the-Art (SOTA) Models

The table below summarises key findings, strengths and limitations of each approach reviewed in the preceding sections.

Table 2.2.1 Comparative summary of SOTA detection methods

Method	Datasets	Results (mAP)	Parameters	FPS/GFLOPS	Key Strengths	Key Limitations
DeC-YOLO (Li et al., 2026)	ExDark, VisDrone, DarkFace, LOL	ExDark: 83.1% VisDrone: 38.2% DarkFace: 44.8%	10.7M (ExDark) 9.3M (VisDrone)	97.6 FPS (ExDark) 97.2 FPS (VisDrone)	Detection gradients enable joint optimisation of brightening and detection	VisDrone Recall of 26.2% means ~74% of targets missed despite high FPS
PFNN (Liu et al., 2025)	VisDrone2019 (Night), Drone Vehicle (Night), ExDark	ExDark: 71.71% mAP@0.5 VisDrone: 23.8% mAP@0.5 Drone Vehicle: 74.9% mAP@0.5	Not reported	40.6 GFLOPs (VisDrone)	Bidirectional gradients prevent over-enhancement by tying enhancement to detection loss	VisDrone Recall of 25.4% means ~74% of high-altitude targets undetected
LDETR (Tiwari et al., 2024)	MS COCO, DAWN (adverse weather), ExDark	MS COCO: 51.8% AP DAWN: 55.85% mAP ExDark: 79.99% mAP	39M	78 GFLOPs	Dual encoder-decoder separates scene structure discovery from object detection	Dust/haze results in 48.38 mAP, a 15.82% drop vs foggy conditions
NF-DETR (Li & Shen, 2025)	BDD100K-Night, City-Night3K	BDD100K-L: AP@50: 58.6% AP@50:95: 82.1% City-Night3K: AP@50: 83.6% AP@50:95: 59.6%	S: 9.8M M: 20.5M L: 31.6M	S: 111.3 FPS / 21.8G M: 86.7 FPS / 61.8G L: 81.7 FPS / 93.4G	FFT-based frequency stream captures illumination-invariant edge and contour cues	Fixed-window FFT unreliable under saturated glare, heavy occlusion, or motion blur

Table 2.2.1 continued: Comparative summary of SOTA detection methods

Method	Datasets	Results (mAP)	Parameters	FPS/GFLOPS	Key Strengths	Key Limitations
DAMFCN + DarkYOLOv8 (Fu et al., 2025)	ExDark, Dark Face	ExDark: 69.8% mAP@0.5 Dark Face: 77.4% mAP	2.9M	6.7 GFLOPs	Lightest model reviewed at 2.9M params and 6.7 GFLOPs is viable for embedded deployment	ExDark Recall of 60.3% means ~40% of objects missed despite high precision
KDFA (Shi et al., 2025)	Custom low-light apple dataset (1,200 pixel-aligned pairs)	mAP@0.5:0.95: 56.1%, +12.7pp vs YOLOv7 baseline	Not reported	Not reported	Geometry-consistent alignment preserves bounding box spatial coherence under distortion	Requires pixel-aligned pairs, which are difficult to acquire in real-world scenes
VLMOE (He et al., 2025)	BDD100k, SHIFT, FLIR	BDD100k: 47.5% mAP@0.5 SHIFT: +1.5% mAP@0.5 vs 2PCNet FLIR: 32.3% mAP@0.5	Not reported	Not reported	CLIP encoder used at training time only, with no nighttime labels required, hence no added inference cost	Teacher is daytime detector, out-of-distribution for nighttime inputs
Debiased Teacher (DeT) (Cui et al., 2025)	BDD100k, SHIFT, TDND	BDD100k: 50.9% mAP@0.5 SHIFT: 52.8% mAP@0.5 TDND: 46.7% mAP@0.5	Not reported	Not reported	ConCal dynamic per-class thresholds replace fixed cutoffs, recovering missed true positives	TDND score (46.7%) trails BDD100k/SHIFT by ~5 mAP, showing sensitivity to severe weather

2.5 Shared Strengths and Cross-Method Limitations

In addition to the method specific limitations mentioned in the preceding sections, the shared strengths and limitations of are explored in this section.

2.5.1 Shared Strengths Across Families

The reviewed methods shared several meaningful advancements, starting from serial enhance-then-detect pipelines, towards joint or coupled optimisation. This includes using shared gradients (Li et al., 2026; Liu et al., 2025), dual-branch transformers (Li & Shen, 2025; Tiwari et al., 2024), or teacher-student distillation (Shi et al., 2025; He et al., 2025; Cui et al., 2025). Ablation studies within each method confirm that adding the joint enhancement or distillation component to a plain detection baseline yields measurable gains.

Another trend is the elimination of hand-crafted post processing. Approaches leveraging RT-DETR (Zhao et al., 2024; Wang et al., 2025; Robinson et al., 2026) remove NMS entirely, avoiding the sensitivity to manually tuned confidence and IoU thresholds that constrain YOLO-based detectors.

Unsupervised and label-efficient strategies are also gaining popularity. Zero-DCE++ in PFNN (Liu et al., 2025) removes the need for paired training data, and VLMOE (He et al., 2025) attempts domain adaptation without any nighttime annotations. These represent practical advantages for real-world deployment where labelled nighttime data is scarce.

2.5.2 Shared Limitations Across Methods

The most consistent quantified failure across reviewed methods is low recall under challenging conditions. DAMFCN/DarkYOLOv8 (Zhao et al., 2025) achieves 60.3% Recall on ExDark, missing approximately four in ten objects, while PFNN (Liu et al., 2025) and DeC-YOLO (Li et al., 2026) both fall to around 25 to 26% Recall on VisDrone

Night, meaning over three quarters of drone-view targets are undetected. These failures occur even in models with strong overall mAP scores, suggesting a bias toward precision over coverage.

Next, most methods are evaluated on a single adverse condition and show meaningful drops when extended. LDETR (Tiwari et al., 2024) drops 15.82% mAP from foggy to dust/haze conditions. NF-DETR (Li & Shen, 2025) acknowledges that its frequency-domain cues fail under saturated glare and severe motion blur.

Category-level generalisation is similarly constrained. Results from DeC-YOLO (Li et al., 2026) span 71.7% gap between its best (car: 86.2%) and worst (awning-tricycle: 14.5%) class on VisDrone. He et al. (2025) showed via t-SNE that visually similar classes, such as bicycle, motorcycle, and rider, remain poorly separated even after domain adaptation.

Finally, assessing the efficiency and accuracy trade-off between the methods remain a challeng, due to a lack of consistency metrics. PFNN (Liu et al., 2025), VLMOE (He et al., 2025), and DeT (Cui et al., 2025) do not report model parameter counts, DAMFCN (Zhao et al., 2025) does not report FPS on its primary benchmark, and KDFA (Shi et al., 2025) omits parameters, GFLOPs, and inference time entirely.

2.6 Gaps in Existing Literature and Motivation for This Study

The review of current literature points towards three opportunities for the present study. First, from an enhancement methods perspective, no existing work examines the effect of image enhancement choices on the annotation quality of a teacher VLM. As such, the relationship between enhancement strategy and VLM pseudo-label quality remains is currently unexplored.

Second, no current work evaluates a structurally unmodified, NMS-free DETR detector as a student model within a low-light knowledge distillation pipeline. This gap is significant given the NMS structural mismatch identified in Section 3.3, which introduces incompatibilities when YOLO-based students are paired with VLM teachers that produce one-to-one pseudo labels.

Third, confidence calibration under degraded inputs remains unaddressed across all reviewed distillation approaches. No study has validated whether teacher confidence scores reliably correspond to prediction quality when the teacher is presented with enhanced or low-light images, which is a fundamental assumption of the knowledge distillation framework that has yet to be examined in this context.

Together, these gaps motivate a study that examines the interaction between low-light enhancement, NMS-free transformer architectures, and teacher VLM pseudo-label quality in an integrated distillation pipeline.

CHAPTER 3: METHODOLOGY

3.1 Benchmark Dataset: ExDark

This research uses the ExDark dataset (Loh & Chan, 2019) as the sole data source. The dataset consists of 7,363 images covering 12 object classes, including Bicycle, Boat, Bottle, Bus and Car, collected from a variety of internet sources and sub-sampled low-light images. The dataset covers 10 distinct low light conditions, ranging from Low to Strong to Twilight, giving the data set a broad level of illumination profiles. Each image is annotated at the image classification and object detection level. The ExDark dataset does not specify a dataset wide train, test and validation split.

The split used in this study is a 80/10/10 stratified split by the primary image class, consistent with the split used by Li et al. (2026) and Zhao et al. (2025). Here, 80% of images are allocated to the training set and 10% to the validation set with integer truncation, with the remaining images allocated to the test set. This yields 5,886 training, 729 validation and 748 test images.

3.2 Proposed Detection Pipeline

The proposed pipeline operates across 3 sequential stages to train a low light real time time DETR. The pipeline overview is illustrated in Figure 3.1.

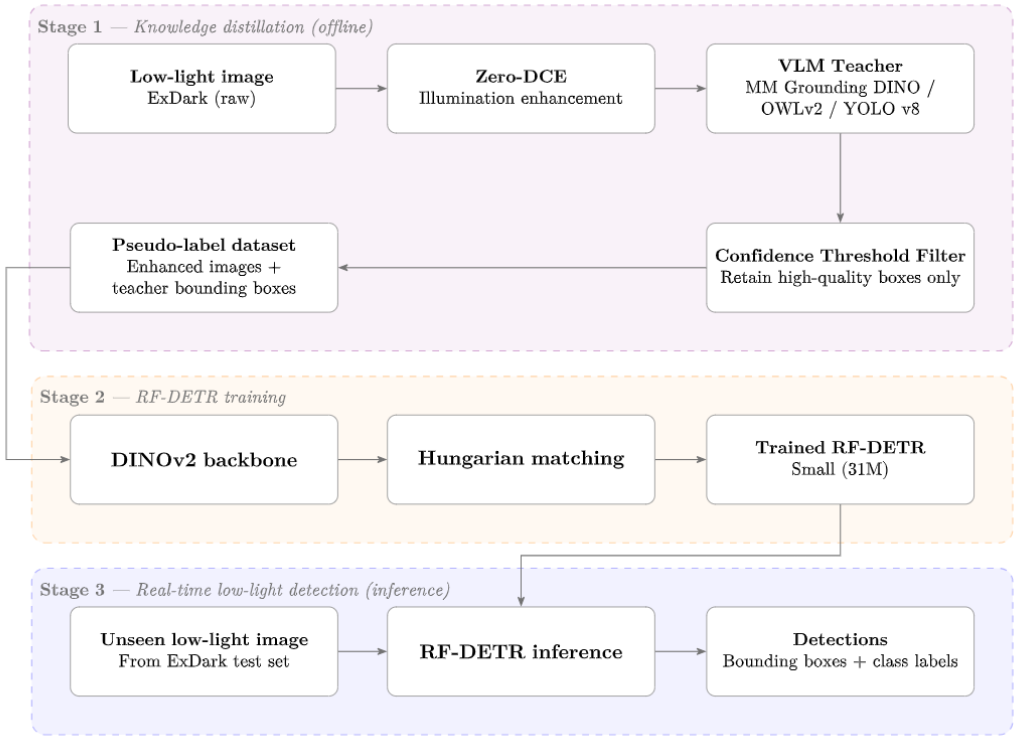


Figure 3.3.1 Detection pipeline architecture

3.2.1 Stage 1: Offline knowledge distillation

At the initial stage, a raw low-light image from the ExDark dataset (Loh & Chan, 2019) is first passed through Zero-DCE (Zero-Reference Deep Curve Estimation) (Guo et al., 2020), an unsupervised illumination enhancement model that adjusts the brightness distribution of the image without requiring paired training data. The enhanced image is then fed to a Vision-Language Model teacher, which is either MM Grounding DINO (Zhao et al., 2024), OWLv2 (Minderer et al., 2023), or YOLO-World v8 (Cheng et al., 2024) that is selected empirically.

The selection criteria for the teacher model are as follows. First, the selection is restricted to fully open-source models with publicly available weights on Hugging Face. Second, the model must be able to produce per-bounding box confidence scores from free-text class-name prompts. As such, API models such as Grounding DINO 1.5 Pro,

and generative VLMs that do not produce calibrated bounding box confidence scores, such as Qwen2.5-VL are explicitly excluded.

The teacher performs zero-shot open-vocabulary object detection, generating bounding box predictions with associated confidence scores for each of ExDark's 12 object classes. These predictions are then passed through a confidence threshold filter, retaining only boxes that exceed a justified, per-class threshold. The surviving high-confidence boxes become the pseudo-labels that when paired with their corresponding enhanced images, form the pseudo-label training dataset. The VLM teacher is used exclusively at this offline annotation stage only and plays no role at inference time.

Given the large-scale pre-training of the VLM teacher models, including images of various lighting conditions, we expect the model representations to be unrestricted to daylight conditions only. Critically, both models produce per-box confidence scores from dedicated detection heads, so high-confidence predictions represent genuine geometric and semantic certainty. When filtered at a high per-class threshold, the retained pseudo-labels are therefore expected to be reliable annotations. The expectation is therefore that the VLM pseudo-labels raise the baseline provided by RT-DETR’s DINO v2 backbone, by adapting the detection head to the target low-light domain.

3.2.2 Stage 2: RF-DETR training

Once the pseudo-label dataset is completed, it is used to train RF-DETR (Robinson et al., 2026) using a standard supervised learning setup. This specific model was selected due to its DINOv2 backbone, which is pretrained on millions of internet images, is expected to provide richer representations for out-of-distribution low-light domains.

During training, RF-DETR's Hungarian bipartite matching loss assigns exactly one predicted bounding box to each pseudo-label ground truth, which is structurally

compatible with the one-to-one nature of VLM-generated pseudo-labels and eliminates the need for NMS. Training is performed on the Small variant (RFDETRSmall, 29M parameters) to mitigate large hardware requirements.

3.2.3 Stage 3: Real-time low-light detection inference

In Stage 3, the trained RF-DETR model is deployed for inference on unseen low-light images. The test images are passed directly through the model end-to-end without any pre-processing or post-processing steps, and the results are analysed and benchmarked against SOTA models on ExDark performance.

3.3 Research Methodology

This research is conducted across the 4 key phases below.

3.3.1 Preliminary stage: VLM teacher selection & threshold analysis

Before any annotation run begins, the VLM teacher model is first empirically selected, and a threshold analysis is conducted to determine a suitable confidence threshold to apply.

First, zero shot inference is performed using all three candidate teachers: MM Grounding DINO, OWLv2 and YOLO-World v8, over a small, stratified subset (600 images) of the ExDark dataset, covering all 12 classes. This is performed on both the raw and Zero-DCE enhanced versions of the training dataset, and the following metrics are computed per class:

1. Confidence score distribution
2. Precision at thresholds: 0.55, 0.60, 0.65, 0.70, 0.75, 0.80. (measured against ExDark ground truth at IoU > = 0.5)
3. Number of images retaining at least one bounding box at each threshold
4. Per-class F1 score

The combination of teacher model, condition and confidence threshold that maximises precision while retaining a significant number of bounding boxes overall and per class, is selected as the final teacher. The hyperparameters and configurations of this stage are summarised below.

Table 3.1 Preliminary stage hyperparameters and configurations

Hyperparameter/ Configuration	**Value**
Random seed	42
Stratified sample	50 images per class x 12 classes = 600 images, with 1,996 ground truth boxes
MM Grounding DINO model ID	openmmlab-community/mm_grounding_dino_tiny_o365v1_goldg_v3det (Hugging Face)
OWLv2 model id	google/owlv2-base-patch16-ensemble (Hugging Face)
YOLO-World v8 weights	yolov8x-worldv2.pt (Ultralytics v8.4.0 release asset)
Zero-DCE weights	Epoch99.pth from Zero-DCE repository
IoU match threshold	0.50
Inference threshold	0.05
Threshold range	0.55, 0.60, 0.65, 0.70, 0.75, 0.80
Minimum surviving bounding boxes per class	10 bounding boxes per class, minimum of 1,000 bounding boxes retained total

The candidate teachers are open-vocabulary teachers that receive object labels as free text and detect what the labels describe. Each of ExDark's 12 classes are therefore given to the models as a text prompt. Here, the classes 'motorbike' and 'people' are renamed to 'motorcycle' and 'person' respectively, as they are expected to be generic wordings more commonly used by common datasets.

The models are prompted in two different ways. MM Grounding DINO and OWLv2 receive all 12 prompts at inference time, once per image. Meanwhile, YOLO-World accepts the 12 prompts once before the entire run begins, converting them into fixed internal representations that are reused for every image. This enables YOLO-World to perform the annotations at a higher rate.

3.3.2 Stage 1: Data engineering and pseudo-label generation

ExDark images are enhanced using Zero-DCE, with the outputs saved. The original ExDark images are retained for image enhancer ablation comparison.

Using the VLM teacher and threshold condition selected at the preliminary phase, a zero shot inferencing run is performed through the ExDark training split. The output bounding boxes are filtered using the determined threshold. A dataset characterisation report is produced to document the class-level survival rate, bounding box count per class, and confidence distribution plots.

3.3.3 Stage 2: Knowledge distillation & training

Structurally identical RF-DETR instances are trained using the native Roboflow rfdetr Python package, using the same hyperparameters. The experimental conditions set up for this study are as follows:

- **Condition A (No intervention baseline):**
    - Trained on raw ExDark images + human-annotated ground-truth labels
    - Validated on raw images.
    - Serves as performance floor of the pipeline
- **Condition B (Spatial enhancement only baseline):**
    - Zero-DCE enhanced images + human annotated ground-truth labels
    - Validated on enhanced images.

    - This benchmark isolates the contribution of spatial enhancement specifically

- **Condition C (Enhanced knowledge distillation):**
    - Zero-DCE enhanced images + VLM pseudo labels
    - Validated on enhanced images
    - Main experimental condition. Tests whether teacher derived pseudo-labels act as a semantic regulariser against enhancement-induced noise, producing a better-trained model than human labels on the same enhanced images
    - Two instances of Condition C were trained, using the confidence threshold of 0.55 and 0.60 respectively, with the justification further explained in the Results section. Empty images with no annotations preserved within the image are dropped from training.

In addition to the three main experimental conditions, an additional phase is conducted to evaluate how the annotation gap between the pseudo-label and human-label trained models can be closed by including a small fraction of human labels within the training dataset.

Two additional conditions: H10 and H25, replace 10% and 25% of the pseudo-labelled training images with their human-labelled equivalent, respectively, using stratified sampling so each class contributes proportionally to the human subset. Both models are trained under the identical configuration and evaluated on the same test split.

- Condition H10:
    - 10% of human labels + 90% pseudo-labels
- Condition H25:
    - 25% of human labels + 75% pseudo-labels

The training configurations and hyperparameters are summarised in the Table 3.2.

Table 3.2 Stage 2 hyperparameters and configurations

Hyperparameter/ Configuration	**Value**
Random seed	42
Student model	RFDETRSmall (rfdetr package), COCO-pretrained initialisation
Epochs	25 with early stopping
Learning rate	1e-4
Batch size	16
Gradient accumulation step	1

Additionally, a zero-shot RF-DERTR run is performed on both raw and enhanced images as a performance comparison floor. The full set of conditions are shown in Table X.

Table 3.3 Full list of RF-DETR training conditions

Condition	**Details**
A	Raw human-labels
B	Enhanced human-labels
C@55	Enhanced pseudo labels, with bounding box confidence threshold of 0.55
C@60	Enhanced pseudo labels, with bounding box confidence threshold of 0.60
H10	Stratified split 90% enhanced pseudo labels + 10% human labels
H25	Stratified split 75% enhanced pseudo labels + 25% human labels
Zero Shot	Zero-shot inference on raw and enhanced images

3.3.4 Stage 3: Comparative benchmarking & analysis

The findings from Stage 2 are comprehensively evaluated, where several outcomes exist within the results space. Potential outcomes and their interpretations are described in Table 3.4 below.

Table 3.4 Potential outcomes and interpretations

Outcome Scenario	Interpretation and next steps
C > B > A	The ideal outcome, where both enhancement and VLM pseudo labels contribute positively and additively. The full pipeline is validated.
B > A, C	Enhancement improves results, but knowledge distillation hurts or does not improve results. This indicates insufficient VLM annotation quality.
C ~= B, C > A	Enhance images drive performance gain, and VLM pseudo labels contributions are equivalent to human data. This means VLM labels can replace human labels while maintaining accuracy.

The final quantitative results, including mAP@0.5, mAP@[.5:.95], per-class AP, parameter count, GFLOPs and frames-per-second are recorded, and compared against results from reviewed literature.

3.3.5 Supplementary study: image enhancer robustness probe

To validate the effect of Zero DCE as the image enhancer, a small, stratified probe is conducted to compare three enhancement conditions:

- Zero DCE: The default image enhancement method.
- CLAHE (Contrast Limited Adaptive Histogram Equalization) (Zuiderveld, 1994): Increases contracts by splitting image into grids and increasing contrast inside each tile.
- SCI (Self-Calibrated Illumination) (Ma et al., 2022): Retinex method that estimates the scene's lighting map and divides it out to recover the underlying surface.

The comparison is conducted across 3 random seeds per configuration, using a 900-image stratified subsample (480 train / 120 validation / 300 test). A total of 12 short training runs, including the raw image condition are conducted, using identical training hyperparameters below.

Table 3.5 Image enhancement robustness probe hyperparameters and configurations

Hyperparameter/ Configuration	**Value**
Random seed/number of runs	0, 1, 2 (4 configurations x 3 seeds = 12 runs total)
Experimental conditions	Raw, Zero-DCE, CLAHE, SCI
Zero-DCE	Epoch99.pth weights
CLAHE	OpenCV createCLAHE, clipLimit = 2.0, tileGridSize = 8×8, applied in LAB colour space to the L (lightness) channel only
SCI	medium.pt weights (trained on LOL + LSRW), inference-only. Only the enhance.* illumination-estimator weights are loaded, the calibrate.* self-calibration branch (training-only) is dropped
Probe subsample	40 / 10 / 25 images per class for train / val / test (480 / 120 / 300 images). 800 images total
Epochs	10 per run, no early stopping
Learning rate	1e-4

An additional step is conducted to investigate the hypothesis of whether enhancement reveals images missing from ground truth and thus receives inaccurate penalisation. The detection of the seed-0 models across the test images were analysed.

Detections are considered as surfaced if they are not already present in the ground truth and are also missed by the control raw detection pipeline. The parameters of the surfaced detections are as below.

Table 3.6 Parameters of surfaced detections

Parameter	Value	Purpose
Confidence floor	0.30	Detection must be more than a marginal guess
Ground-truth IoU	below 0.30	No ground truth box covers the detection
Cluster IoU	0.50 or above	Detections from different enhancers merged into one object
Consensus	1/3, 2/3 or 3/3	How many enhancers independently surfaced the object
Raw-missed flag	0 or 1	Denotes whether raw model also fails to detect the object
Seed	0 only	One model per condition, not all three

CHAPTER 4: RESULTS

4.1 Preliminary Phase

The Preliminary Phase is conducted as described in Section 3.3.1 to select the best VLM teacher. The candidate models, MM Grounding DINO, OWLv2 and YOLO World v8, are run on both the raw and enhanced versions of every sampled image in zero-shot mode, using each class name as a text prompt.

4.1.1 Confidence separation

The pipeline keeps a pseudo-label only if its confidence score exceeds a threshold. The confidence distributions for correct versus incorrect bounding boxes are plotted below.

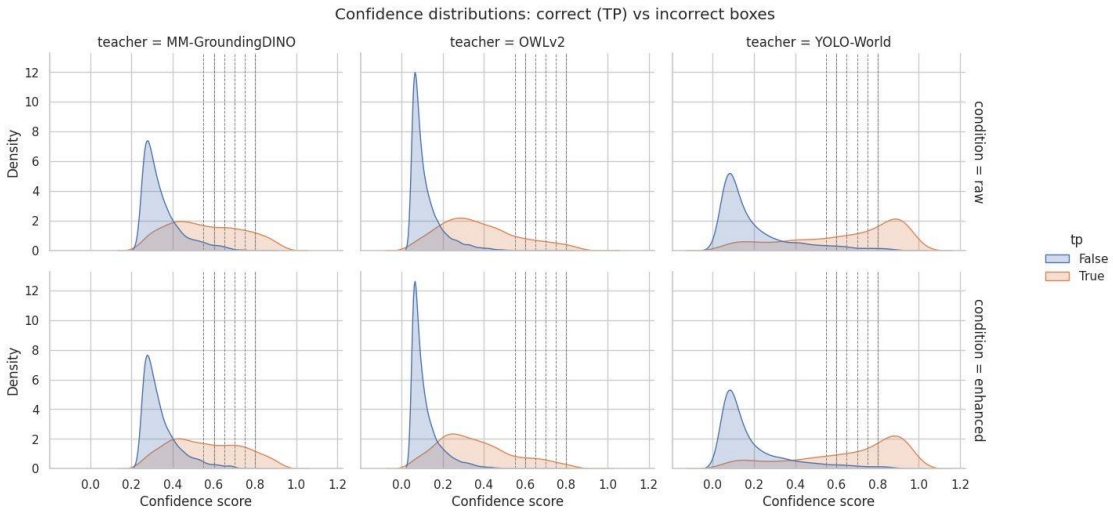


Figure 4.1 Confidence score distributions for correct (True) vs incorrect (False) boxes, by teacher and image condition. Dashed vertical lines mark the candidate thresholds.

YOLO-World gives the best confidence separation, with correct predictions clustered towards high confidence scores, creating a peak at the high confidence region. On the other hand, OWLv2's predictions cluster towards the lower confidence range, at a region of about 0.4 or lower confidence. MM Grounding DINO gives a moderate confidence separation relative to the other 2 models.

4.1.2 Threshold Precision

The precision of retained bounding boxes is evaluated at 6 confidence thresholds, ranging from 0.55 to 0.8, for all combinations of teacher and enhancement conditions.

Table 4.1 Precision of retained boxes at each candidate threshold

Teacher · Condition	0.55	0.60	0.65	0.70	0.75	0.80
MM Grounding DINO · raw	0.810	0.862	0.919	0.955	0.971	0.974
MM Grounding DINO · enhanced	0.819	0.859	0.896	0.951	0.970	0.978
OWLv2 · raw	0.881	0.894	0.910	0.945	0.934	0.976
OWLv2 · enhanced	0.862	0.864	0.900	0.938	0.966	1.000
YOLO-World · raw	0.835	0.858	0.888	0.905	0.920	0.942
YOLO-World · enhanced	0.829	0.849	0.874	0.897	0.912	0.931

4.1.3 Performance comparison

Additionally, the per class F1 score is computed across classes for each teacher and condition combination to verify teacher model performance. A snapshot of per-class F1 at threshold of 0.60 is reported in Table 4.2.

Table 4.2 Per-class F1 score at threshold 0.60

Class	MM-GDINO enh.	MM-GDINO raw	OWLv2 enh.	OWLv2 raw	YOLO-World enh.	YOLO-World raw
Bicycle	0.600	0.640	0.598	0.625	0.763	0.759
Boat	0.339	0.349	0.460	0.538	0.500	0.561
Bottle	0.500	0.476	0.012	0.070	0.618	0.586
Bus	0.815	0.839	0.919	0.930	0.898	0.941
Car	0.594	0.595	0.000	0.000	0.710	0.704
Cat	0.488	0.429	0.778	0.811	0.667	0.696
Chair	0.328	0.306	0.000	0.025	0.620	0.590
Cup	0.426	0.432	0.000	0.015	0.655	0.683
Dog	0.473	0.545	0.648	0.685	0.694	0.703
Motorbike	0.364	0.393	0.500	0.576	0.729	0.729
People	0.544	0.545	0.000	0.000	0.696	0.687
Table	0.305	0.283	0.000	0.013	0.148	0.114
Mean	**0.481**	**0.486**	**0.326**	**0.358**	**0.641**	**0.646**

Overall, YOLO-World leads in 10 of the 12 classes, with the mean of 0.641 and 0.646 for enhanced and raw images, respectively. OWLv2 performs the poorest, achieving zero F1 score on 5 classes on enhanced images: Car, Chair, Cup, People and Table. MM Grounding DINO produced the second-best results, behind YOLO-World.

Table is the weakest class for YOLO-World at 0.148 for enhanced images. Across the board, image enhancement was not beneficial to increasing teacher model performance.

Figure 4.2 visualises an example output for all 6 cases. Note that Owlv2 produces many additional annotations at 26 versus 4 ground truth boxes.

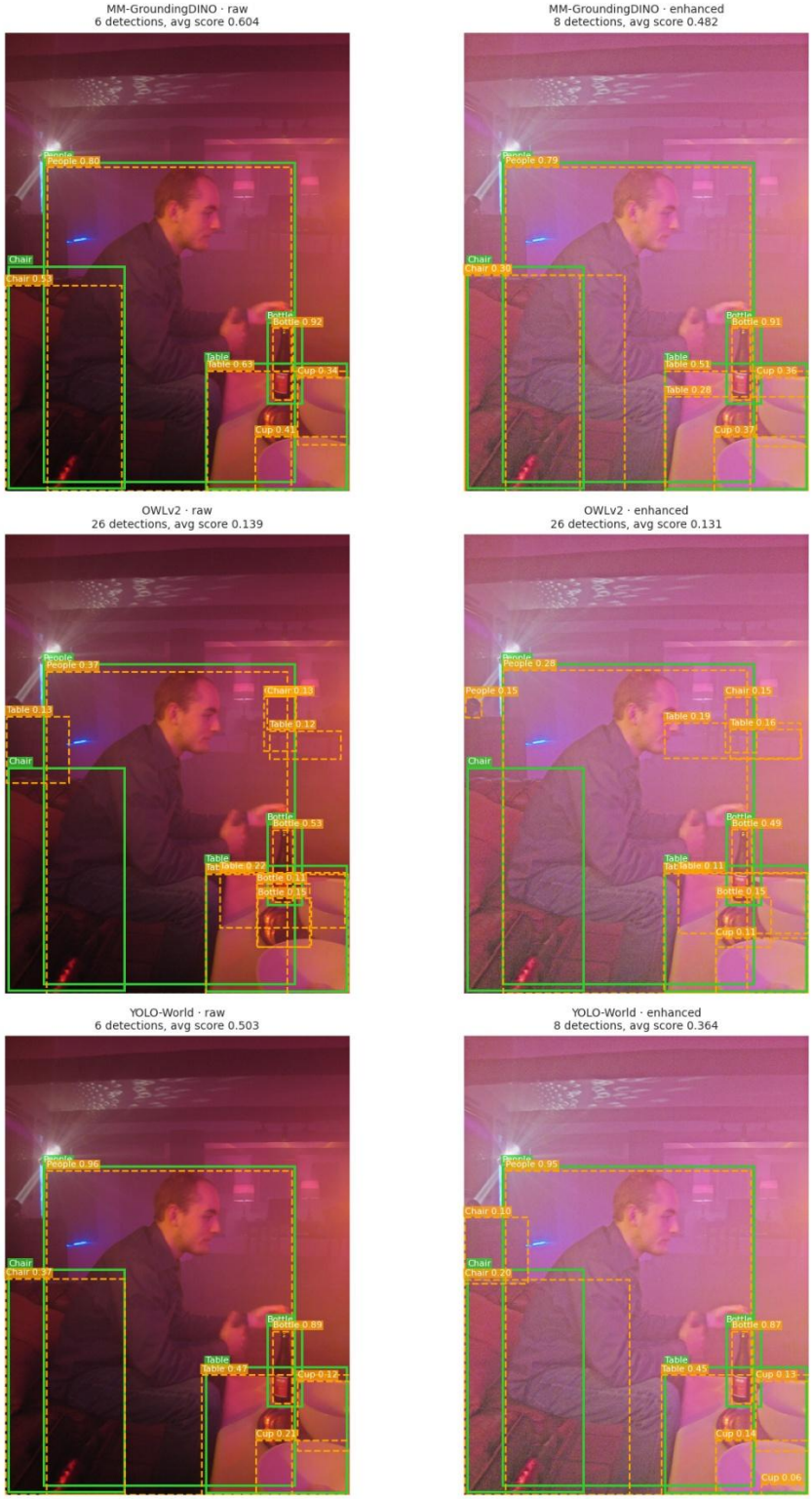


Figure 4.2 Ground truth (solid green) vs. top 10 predictions (dashed orange) for one sample image, across all six teacher-condition combinations.

4.1.4 Teacher selection

Precision alone is insufficient as a criterion for teacher selection, as precision for a particular model can be raised by keeping fewer bounding boxes, such that the dataset becomes too small to train on. Table 4.4 below shows this across the 3 models under enhanced conditions and confidence thresholds of 0.60. Here, OWLv2 gives the highest precision, but only keeps a small number of 265 boxes versus the ground truth count of 1,996.

Table 4.3 Teacher precision, boxes kept, and mean per class F1 performance at 0.60 confidence, under enhanced condition

Teacher	Precision	Boxes kept	Classes with zero F1	Mean per-class F1
OWLv2	0.864	265	5 of 12	0.326
MM Grounding DINO	0.859	802	0 of 12	0.481
YOLO-World	0.849	1,255	0 of 12	0.641

As such, 2 additional constraints are introduced, where the final teacher and condition must retain at least 1,000 boxes overall, corresponding to about 50% of the original ground truth count, which rules out sparse label sets, and at least 10 boxes in every one of the 12 classes, which rules out label sets that are dense overall but empty for particular classes.

5 Teacher-condition combinations fulfil these criteria. The top performer, YOLO-World with enhanced condition, at confidence threshold of 0.60. This threshold is further tested in the full pseudo label generation step of Phase 1.

Table 4.4 Teacher, condition and threshold combinations that fitful teacher selection criteria

Teacher	Condition	Threshold	Precision	Boxes kept
YOLO-World	enhanced	0.60	0.849	1,255
YOLO-World	raw	0.55	0.835	1,342
YOLO-World	enhanced	0.55	0.828	1,371
MM Grounding DINO	enhanced	0.55	0.819	1,008
MM Grounding DINO	raw	0.55	0.810	1,016

4.2 Stage 1: Data engineering and pseudo-label generation

This stage produces the full pseudo-label dataset for Stage 2, while validating the operating point from the Preliminary phase. The previously selected teacher, YOLO-World V8, is run across the entire training split on both raw and enhanced copies across the entire confidence threshold range.

4.2.1 Precision, recall and F1 for raw and enhanced conditions

Pseudo-label performance is evaluated across the full confidence threshold, for the full dataset.

Table 4.5 Pseudo-label precision, recall and F1 on the full training split, raw vs enhanced

Threshold	Raw Prec.	Raw Rec.	Raw F1	Enh. Prec.	Enh. Rec.	Enh. F1	F1 Change
0.55	0.829	0.567	0.673	0.819	0.573	**0.674**	+0.001
0.60	0.851	0.532	0.655	0.843	0.539	**0.657**	+0.003
0.65	0.875	0.494	0.632	0.867	0.503	**0.637**	+0.005
0.70	0.897	0.452	0.601	0.890	0.458	**0.605**	+0.004
0.75	0.919	0.402	0.560	0.913	0.410	**0.566**	+0.006

Enhancement wins marginally, though the improvement is minimal across all confidence thresholds. This is further visualised in the performance-threshold, and confidence curves in Figure 4.3.

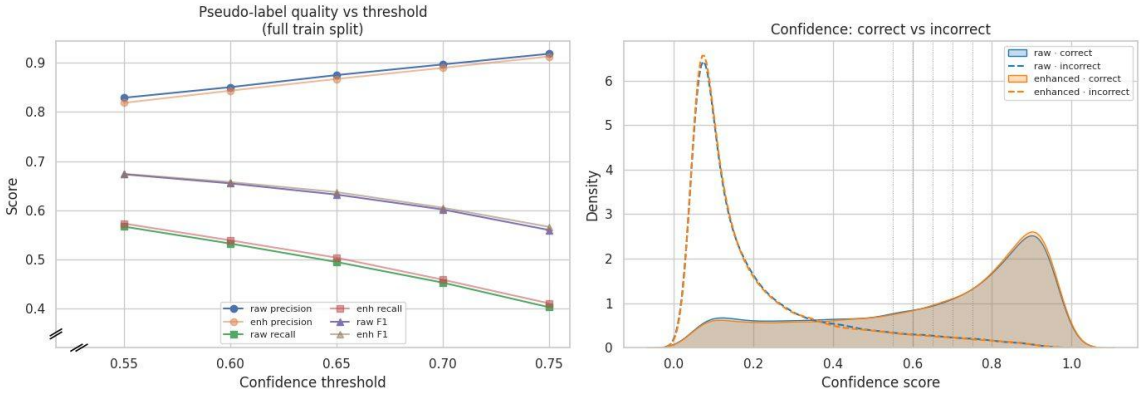


Figure 4.3 Left: precision, recall and F1 against threshold for raw and enhanced pseudo-labels on the full training split. The two sets of curves are nearly coincident at every threshold. Right: confidence distributions for correct and incorrect boxes, for raw versus enhanced.

4.2.2 Threshold refinement and precision-coverage trade-off

Threshold selection determines the balance between precision of labels versus the number of retained labels. As discussed, the optimal threshold cannot be determined by precision alone. Table 4.6 below documents YOLO-World's precision-coverage trade-off for the enhanced condition, which demonstrates the expected behaviour of increased precision, but reduced number of boxes kept as threshold increases.

Table 4.6 Precision-coverage trade-off, YOLO-World enhanced

Threshold	Precision	Recall	F1	Boxes kept	Image coverage	Min. class boxes
0.55	0.819	0.573	0.674	13,366	0.872	131
0.60	0.843	0.539	0.657	12,200	0.852	99
0.65	0.867	0.503	0.637	11,084	0.826	70
0.70	0.890	0.458	0.605	9,842	0.792	36
0.75	0.913	0.410	0.566	8,582	0.749	15

The threshold of 0.55 maximises the F1 score while retaining a high number of annotations, making it a strong candidate threshold for Stage 2. This is treated as a secondary choice that is also carried to the Stage 2 alongside the original candidate threshold of 0.60. Note that a high threshold such as 0.75 retains per-class bounding boxes as low as 15 boxes, making it unsuitable for pseudo-label generation.

4.2.3 Per-class pseudo-label quality

Table 4.7 reports the per class pseudo-label quality for the original candidate threshold. Precision remains stable across all classes. As expected from the Preliminary phase, Table remains the weakest class, with a per-class F1 of 0.124 due to low recall, and a relatively low number of 99 retained boxes, out of which 81 are true positives. In comparison, the largest class, People, has 4,040 number of boxes kept.

Table 4.7 Per-class pseudo-label quality, enhanced condition at threshold 0.60

Class	GT	Boxes kept	TP	Precision	Recall	F1
Bus	559	531	481	0.906	0.860	0.883
Bicycle	909	673	599	0.890	0.659	0.757
Car	2,352	1,674	1,436	0.858	0.611	0.713
Dog	818	644	518	0.804	0.633	0.709
People	5,975	4,040	3,514	0.870	0.588	0.702
Cup	1,352	1,014	800	0.789	0.592	0.676
Motorbike	890	492	461	0.937	0.518	0.667
Cat	719	490	399	0.814	0.555	0.660
Bottle	1,322	956	735	0.769	0.556	0.645
Chair	1,892	1,090	893	0.819	0.472	0.599
Boat	1,113	497	372	0.748	0.334	0.462
Table	1,204	99	81	0.818	0.067	0.124

Decreasing the threshold to 0.55 offers a potential solution. This increases the number of retained boxes by 1,166, including adding 32 bounding boxes to Table. The additional labels are especially beneficial to low coverage classes, such as Boat (+68) and Motorbike (+44).

Table 4.8 Per-class box counts at both thresholds, enhanced condition.

Class	GT boxes	Boxes kept @0.60	Boxes kept @0.55	Gain	Gain %
Table	1,204	**99**	**131**	+32	+32.3%
Boat	1,113	**497**	**565**	+68	+13.7%
Chair	1,892	**1,090**	**1,236**	+146	+13.4%
Bottle	1,322	**956**	**1,063**	+107	+11.2%
Cup	1,352	**1,014**	**1,117**	+103	+10.2%
Car	2,352	**1,674**	**1,829**	+155	+9.3%
People	5,975	**4,040**	**4,416**	+376	+9.3%
Motorbike	890	**492**	**536**	+44	+8.9%
Bicycle	909	**673**	**725**	+52	+7.7%
Dog	818	**644**	**683**	+39	+6.1%
Cat	719	**490**	**516**	+26	+5.3%
Bus	559	**531**	**549**	+18	+3.4%
Total	19,105	**12,200**	**13,366**	+1,166	+9.6%

4.2.4 Final pseudo-label set

The final pseudo-label datasets for A (raw-human), B (enhanced-human). C@0.55 (enhanced-pseudo-label, 0.55 confidence) and C@0.60 (enhanced-pseudo-label, 0.60 confidence) are built and saved. Each set shares the same training, validation and test splits to ensure fair comparisons downstream.

Note that the pseudo-label sets contain images containing 0 annotations, due to confidence filtering. These images are discarded during the training phase, as they do not contribute meaningful labels and may train the detector to bias towards under-detection.

Table 4.9 The two pseudo-label sets carried into Stage 2 (C@0.55, C@0.60), against the human baseline (A and B)

Property	Human labels (A and B)	C@0.55	C@0.60
Training images available	5,886	5,886	5,886
Images carrying at least one box	5,886	5,134	5,012
Images with no label at all	0	752 (12.8%)	874 (14.8%)
Total boxes	19,105	13,366	12,200
Boxes as a share of human	100%	70%	64%
Boxes per labelled image	3.25	2.60	2.43
Smallest class, by box count	559 (Bus)	131 (Table)	99 (Table)

4.3 Stage 2: Knowledge distillation and training

4.3.1 Conditions A, B, C@0.55 and C@0.60

Training is conducted for similar configurations of RF-DETR, across the 4 training conditions (A, B, C@0.55, C@0.60) using the hyperparameters as described in Section 3.3.3. Note that RF-DETR's pipeline includes a default data augmentation stack of resize/crop scale jitter, plus a horizontal flip at 50%, that is applied to the training data before training begins. All variations converged successfully, as shown by the loss and validation mAP curves in Figure 4.

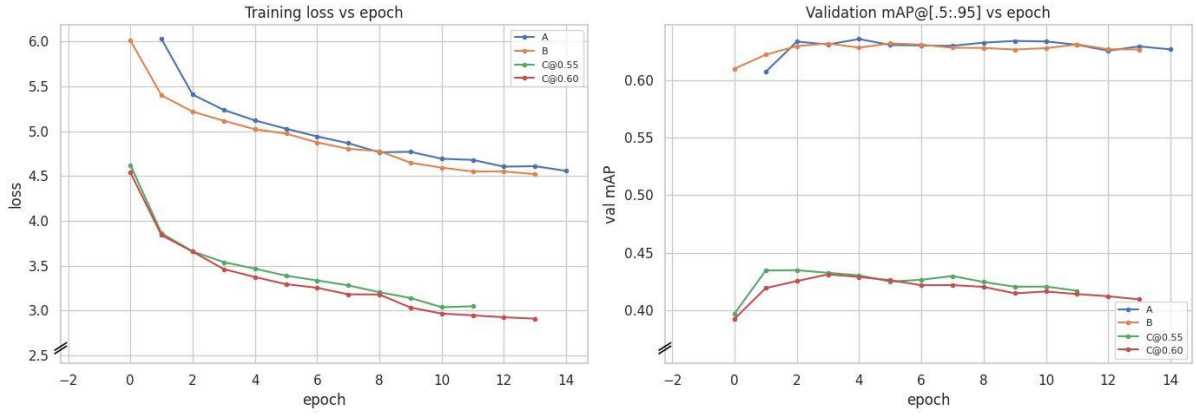


Figure 4.4 Training loss (left) and validation mAP@[.5:.95] (right) per epoch. The pseudo-label conditions converge to a lower training loss than the human-labelled conditions while reaching a lower validation mAP.

Table 4.10 Training results for 4 experimental conditions

Condition	Description	mAP@0.5	mAP@[.5:.95]	Tightness
A	Raw + human labels	0.893	0.634	0.710
B	Enhanced + human labels	0.883	0.621	0.703
C@0.55	Enhanced + pseudo-labels	0.758	0.420	0.554
C@0.60	Enhanced + pseudo-labels	0.755	0.416	0.551

The results for all 4 conditions are recorded in table X. Overall, human-labelled conditions remain superior, producing an mAP@0.5 of 0.893 and mAP@[.5:.95] of 0.634. Enhancement was again found to not add any meaningful contribution to the results, with condition B giving a slightly lower performance than raw performance labels.

C@0.55 exceeds C@0.60 by 0.003-0.004 in mAP, meaning the lower threshold and its larger, noisier label set produced the slightly better model. C@0.55 is retained as the pseudo-label model for further exploration.

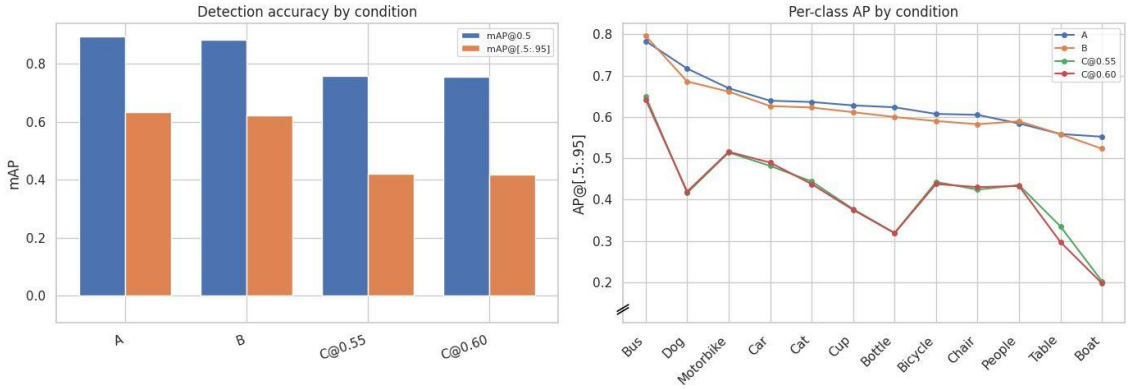


Figure 4.5 mAP@0.5 and mAP@[.5:.95] by condition (Left). Per-class AP@[.5:.95], classes ordered by Condition A performance (Right). The gap between the human-labelled and pseudo-label pairs holds across every class. A detailed per-class comparison is conducted in Section 4.4.

4.3.2 Hybrid pseudo-label conditions, H10 and H25

Using C@055 as base, H10 and H125 are trained by replacing 10% and 25% of the pseudo-labelled training images with their human-labelled equivalents. Selection is stratified by class, so each class contributes to the human subset in proportion to its size.

Table 4.11 Hybrid dataset composition

Condition	Human images	Pseudo images	Human bounding boxes	Pseudo bounding boxes	Total
H10	589	4,618	1,865	12,069	5,207 img / 13,934 box
H25	1,472	3,852	4,700	10,095	5,324 img / 14,795 box

Table 4.12 Hybrid pseudo-label condition results

Human label %	Condition	Human images	mAP @0.5	mAP @[.5:.95]	Gap recovery mAP@0.5	Gap recovery mAP@[.5:.95]
0%	C@0.55	0	0.758	0.420	0.0%	0.0%
10%	**H10**	**589**	**0.799**	**0.449**	**33.0%**	**14.5%**
25%	**H25**	**1,472**	**0.827**	**0.479**	**55.0%**	**29.5%**
100%	B	5,886	0.883	0.621	100%	100%

The addition of human labels proves to be beneficial to mAP. Just 10% of human labels closes the performance gap of mAP@0.5 by 33%. Adding 25% of human labels closes this gap by 55%. This effect is less pronounced for mAP@[.5:.95], with 14.5% and 29.5% of the gap recovered for 10% and 25% of human labels respectively.

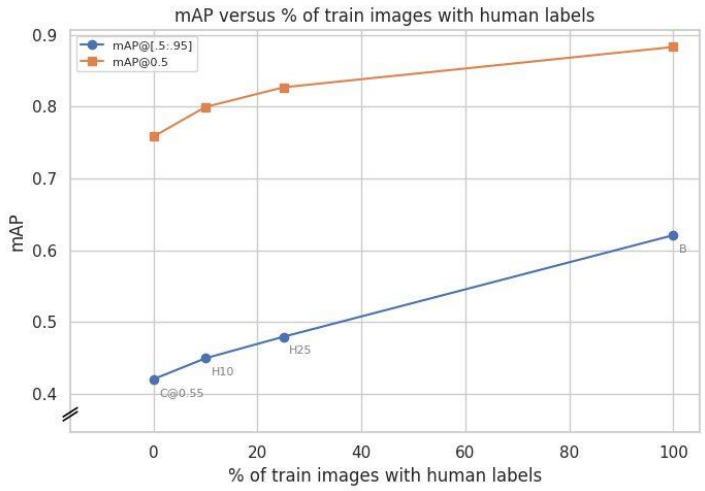


Figure 4.6 mAP against the percentage of training images carrying human labels, both mAP. The mAP@0.5 curve rises moderately steeply over the first quarter of the budget and then flattens, while the mAP@[.5:.95] curve is close to linear.

A full comparison of per class performance is conducted in the following section, across all training conditions.

Table 4.13 Comparative per class performance of hybrid labels versus full pseudo-labels and human labels

Class	C@0.55	H10	H25	A	Gap recovery H10	Gap recovery H25
Table	**0.335**	**0.435**	**0.479**	**0.558**	**44.6%**	**64.5%**
Bicycle	**0.443**	**0.448**	**0.502**	**0.590**	**3.1%**	**40.4%**
Chair	**0.424**	**0.437**	**0.486**	**0.583**	**8.0%**	**38.7%**
Motorbike	0.514	0.550	0.568	0.661	24.1%	36.3%
Car	0.482	0.507	0.530	0.627	17.8%	33.5%
Boat	0.203	0.258	0.300	0.523	17.2%	30.3%
Cup	0.377	0.401	0.438	0.612	10.0%	25.8%
Bus	**0.650**	**0.658**	**0.686**	**0.796**	**5.2%**	**24.8%**
People	0.435	0.452	0.471	0.590	10.6%	22.8%
Cat	**0.444**	**0.458**	**0.483**	**0.623**	**7.5%**	**21.8%**
Dog	0.416	0.451	0.454	0.686	12.8%	14.1%
Bottle	0.319	0.341	0.356	0.600	7.6%	13.1%

4.4 Stage 3: Comparative benchmarking and analysis

4.4.1 Comparison across all conditions with a zero-shot baseline

Results across all training conditions were compared with a COCO-pretrained RF-DETR-Small, which acts as a zero-shot baseline for this study. The outputs of this model are restricted to the 12 ExDark classes, for both raw and enhanced cases. The results are summarised below.

Table 4.14 Performance comparison of all conditions versus baseline

Run	Images	Trained	mAP@0.5	mAP@[.5:.95]
A	raw	yes	**0.893**	**0.634**
B	enhanced	yes	0.883	0.621
H25	enhanced	yes	0.827	0.479
H10	enhanced	yes	0.799	0.449
C@0.55	enhanced	yes	0.758	0.420
C@0.60	enhanced	yes	0.755	0.416
Zero-shot	raw	no	0.732	0.402
Zero-shot	enhanced	no	0.729	0.402

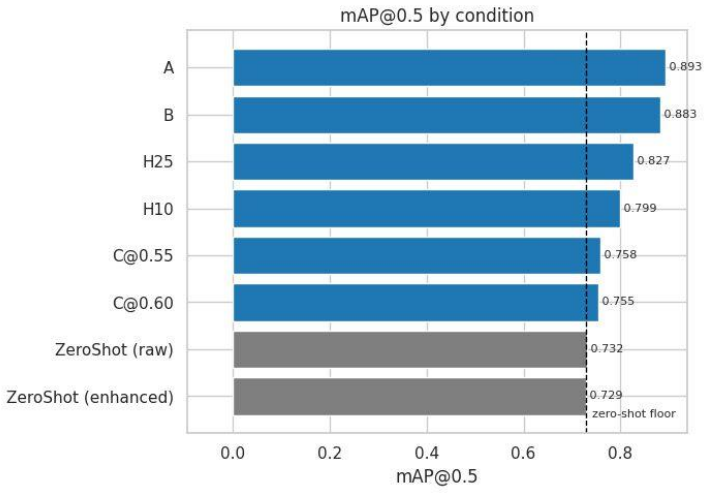


Figure 4.7 mAP@0.5 comparison of all conditions versus baseline

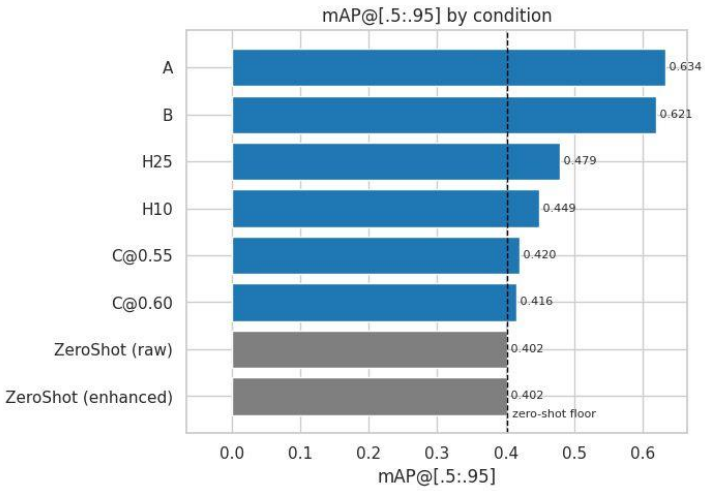


Figure 4.8 mAP@[.5:.95] comparison of all conditions versus baseline

RF-DETR is found to produce strong results out of the box, with 0.732 mAP@0.5 and 0.402 mAP@[0.5:0.95]. Pure pseudo-labels add small improvements to both mAPs, with C@0.55 improving mAP@0.5 by 0.026 and mAP@[0.5:0.95] by 0.019.

4.4.2 Per-class performance against zero-shot baseline

The per-class APs for all configurations are compared versus the baseline.



Figure 4.9 Per-class AP@0.5 zero-shot baseline versus trained conditions

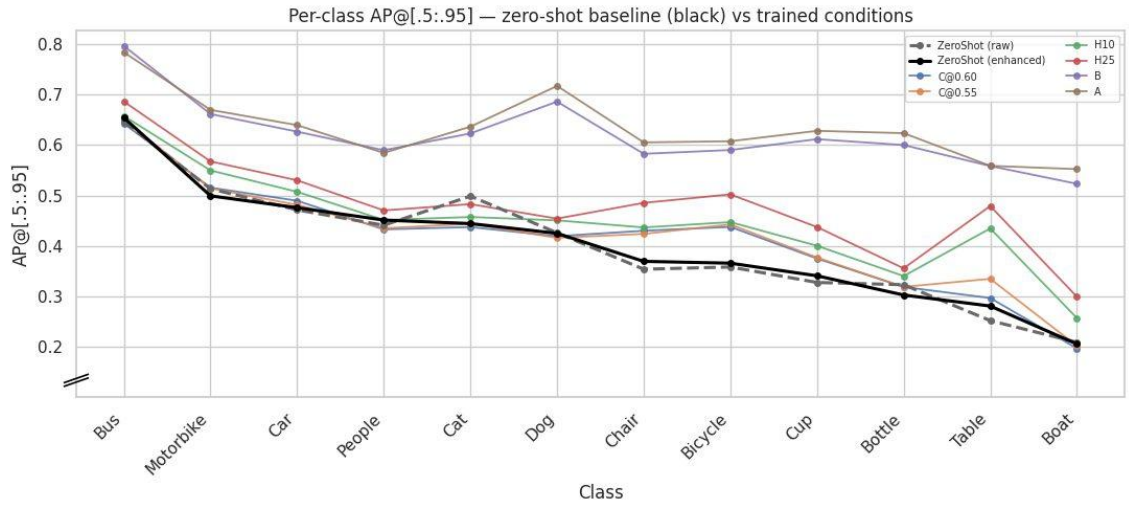


Figure 4.10 Per-class AP@[.5:.95] zero-shot baseline versus trained conditions

The per class AP values are tabulated below, sorted from zero-shot up to the performance ceiling, i.e. condition A.

Table 4.15 Per-class AP@0.5

Class	**ZS raw**	**ZS enh.**	**C@0.55**	**H10**	**H25**	**B**	**A**
Bus	0.927	0.930	0.929	0.927	0.946	0.943	0.940
Motorbike	0.852	0.874	0.843	0.887	0.888	0.910	0.927
Car	0.819	0.817	0.815	0.844	0.857	0.896	0.910
People	0.818	0.816	0.803	0.823	0.843	0.902	0.897
Cat	0.834	0.783	0.776	0.817	0.848	0.926	0.918
Dog	0.816	0.818	0.851	0.854	0.873	0.921	0.931
Chair	0.655	0.678	0.760	0.784	0.810	0.848	0.876
Bicycle	0.720	0.681	0.819	0.820	0.874	0.878	0.877
Cup	0.649	0.666	0.732	0.758	0.786	0.866	0.878
Bottle	0.698	0.690	0.725	0.761	0.770	0.868	0.889
Table	0.385	0.439	0.501	0.658	0.708	0.783	0.792
Boat	0.605	0.555	0.544	0.658	0.717	0.854	0.880

4.4.3 Localisation analysis

Table 4.16 Prediction tightness relative to ExDark ground truth

Condition	Matched detections	% Detections tighter (>= 1.25 GT Area)	% Detections looser (<= 1.25 GT Area)	Median GT/pred area ratio
Zero-shot · raw	1,480	**38.4**	1.2	1.192
Zero-shot · enhanced	1,530	36.3	2.0	1.179
C@0.60	1,264	33.1	2.8	1.161
C@0.55	1,331	32.7	2.8	1.158
H10	1,352	32.3	3.0	1.148
H25	1,364	**27.9**	2.5	1.138
B	1,731	6.2	8.0	0.994
A	1,678	5.9	6.6	0.994

The distribution of boxes tighter than ground truth and visualisation of percentage of annotations tighter than ground truth by condition are captured in Figure 4.10.

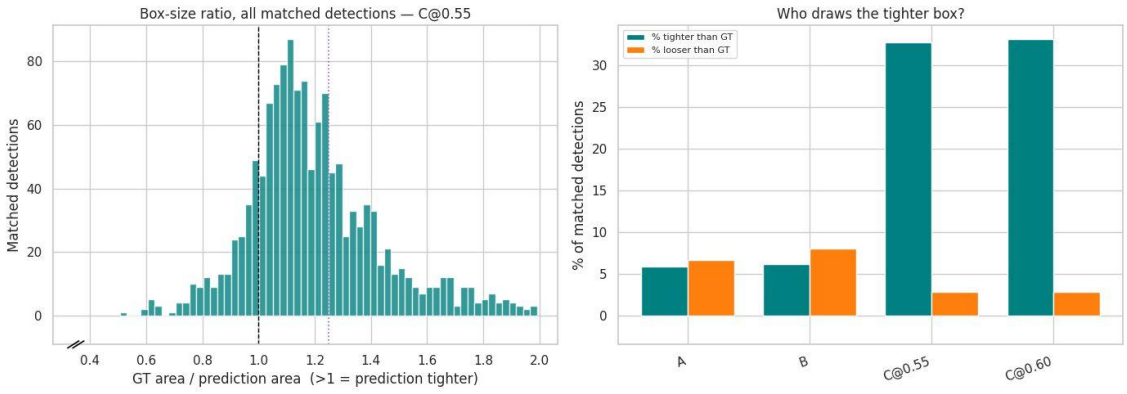


Figure 4.11 Distribution of the ratio of ground-truth area to prediction area for all matched detections under C@0.55 (Left). Instances to the right of 1.0 means the prediction is tighter than the ground truth. Percentage of matched detections tighter and looser than ground truth (Right).

To visualise these cases, Figure 4.11 shows randomly selected cases of predictions that produce tighter annotations as ground truth, with IoU ranging from 0.51 to 0.53. Visual inspection shows that the IoU penalisation may not be truly justified, as the predicted boxes for the cases for Cup, Bottle and People are correctly tighter around the object

boundary. Nonetheless, the Boat example on row 3 demonstrates an overly tight annotation, where the prediction misses the bottom of the boat, possibly because the model mistakes this portion as the boat's reflection.

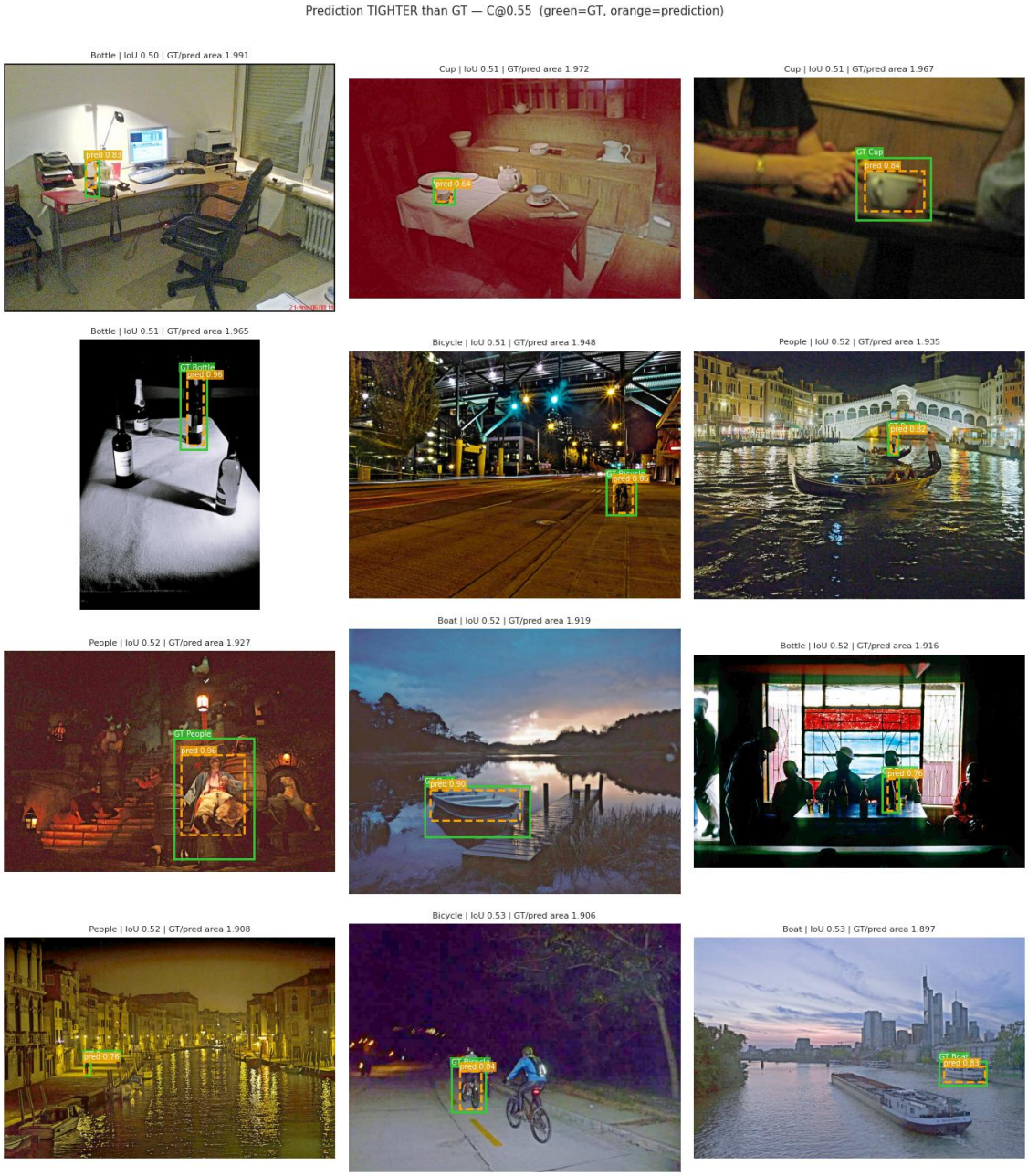


Figure 4.12 Example cases where prediction boxes are tighter than ground truth, for C@0.55

4.4.4 Supplementary study: image enhancer robustness probe

Enhancement was consistently found to not improve detection performance. This supplementary study attempts to validate this claim through a multi-seed evaluation, as

described in Section 3.3.5. Figure 4.22 below provides a snapshot of images enhanced by the 3 illumination methods, next to the raw image.

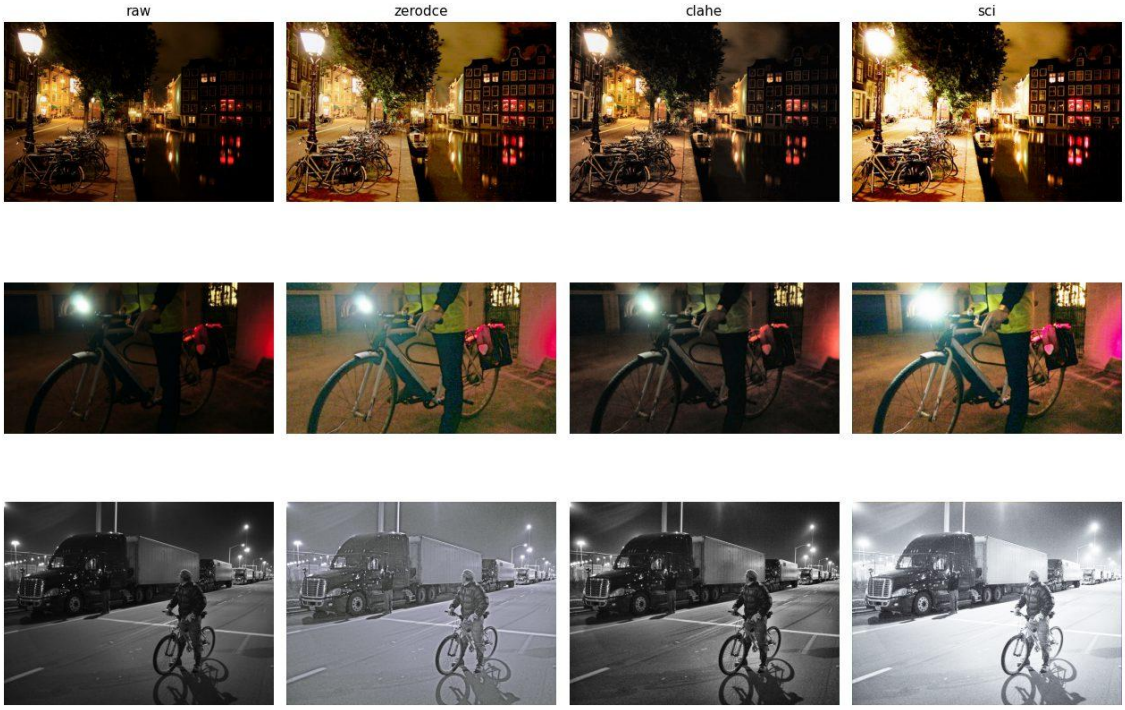


Figure 4.13 Examples images in raw and the 3 selected enhancement conditions

Zero-DCE and SCI brighten the image globally and amplifies sensor noise in the dark regions. True to the nature of histogram equalisation, CLAHE re-distributes pixel brightness to increase contrast, leading to a lower overall brightness shift.

Table 4.17 Probe results across 3 seeds, ordered by mean mAP@[.5:.95]

Condition	Seeds	mAP@0.5	mAP@[.5:.95]	± Std mAP@[.5:.95]	Per-seed values mAP@[.5:.95]	Tightness
raw	3	**0.8380**	**0.5722**	0.0029	0.5689, 0.5739, 0.5738	0.683
CLAHE	3	0.8346	0.5677	0.0037	0.5712, 0.5682, 0.5638	0.680
SCI	3	0.8295	0.5644	0.0073	0.5563, 0.5703, 0.5666	0.680
Zero-DCE	**3**	**0.8252**	**0.5631**	**0.0073**	**0.5547, 0.5664, 0.5682**	**0.682**

The results across the 3 enhancement methods confirm the impact of enhancement. All methods, across all seeds, were found to marginally degrade the raw mAP. Zero-DCE retains the tightness score that is closest to the raw image performance.

Table 4.18 Surfaced objects by consensus and confidence

Consensus	**0.30-0.50**	**0.50-0.70**	**0.70-0.90**	**0.90-1.00**	**Total**	**% of GT**
3 of 3 methods	21	41	22	1	**85**	9.6%
2 of 3 methods	38	15	4	0	57	6.4%
1 of 3 methods	126	12	1	0	**139**	15.7%
Total	**185**	**68**	**27**	**1**	**281**	**31.6%**

Table 4.17 summarises the number of surfaced objects detected from the probe, that were not found in the ground truth labels. The scan found 281 distinct surfaced objects across the 300 test images, equal to 31.6% of the 888 ground-truth boxes in that split, which is a significant number. Objects found by only one method sit almost entirely in the lowest confidence band, 126 of 139 objects, which suggest the profile of noise rather than discovery. Objects found by all three methods are spread far more evenly across confidence, with 64 of 85 objects above 0.50 confidence, which suggests genuinely real objects.

Table 4.19 Surfaced objects that the raw model also missed, broken down by consensus and confidence.

Consensus	**0.30-0.50**	**0.50-0.70**	**0.70-0.90**	**0.90-1.00**	**Raw-missed**	**Of surfaced**	**% of GT**
3 of 3 methods	6	5	0	0	**11**	**12.9%**	1.2%
2 of 3 methods	19	3	0	0	22	38.6%	2.5%
1 of 3 methods	**93**	7	0	0	**100**	**71.9%**	11.3%
Total	**118**	**15**	**0**	**0**	**133**	**47.3%**	**15.0%**

Out of the 281 newly surfaced labels, 118 are missed by the control raw condition. These represent new labels surfaced due to enhancements, as shown in Table 4.18. Overall, majority, 104 out of 120 images lie in the low confidence range, suggesting that most of the detections are potentially noise versus genuine new detections.



Figure 4.14 Example of a correct surface detection (Bicycle, bottom right), detected after enhancement



Figure 4.15 Example of incorrect surfaced detection of the label Chair. Note that trains are not part of ExDark's labels.

In that sense, there is no conclusive evidence that enhancement produces a significant number of new detections that can challenge the validity of the ExDark ground truth, without a full analysis of the ground truth labels. The figures below document samples of surfaced detections with consensus from 1,2 and 3 enhancement methods respectively. While some detections are incorrect, other samples demonstrate genuine new detections that can increase the model's mAP when accounted for appropriately. This is because correct detections missed by ground truth lead to mAP penalties.

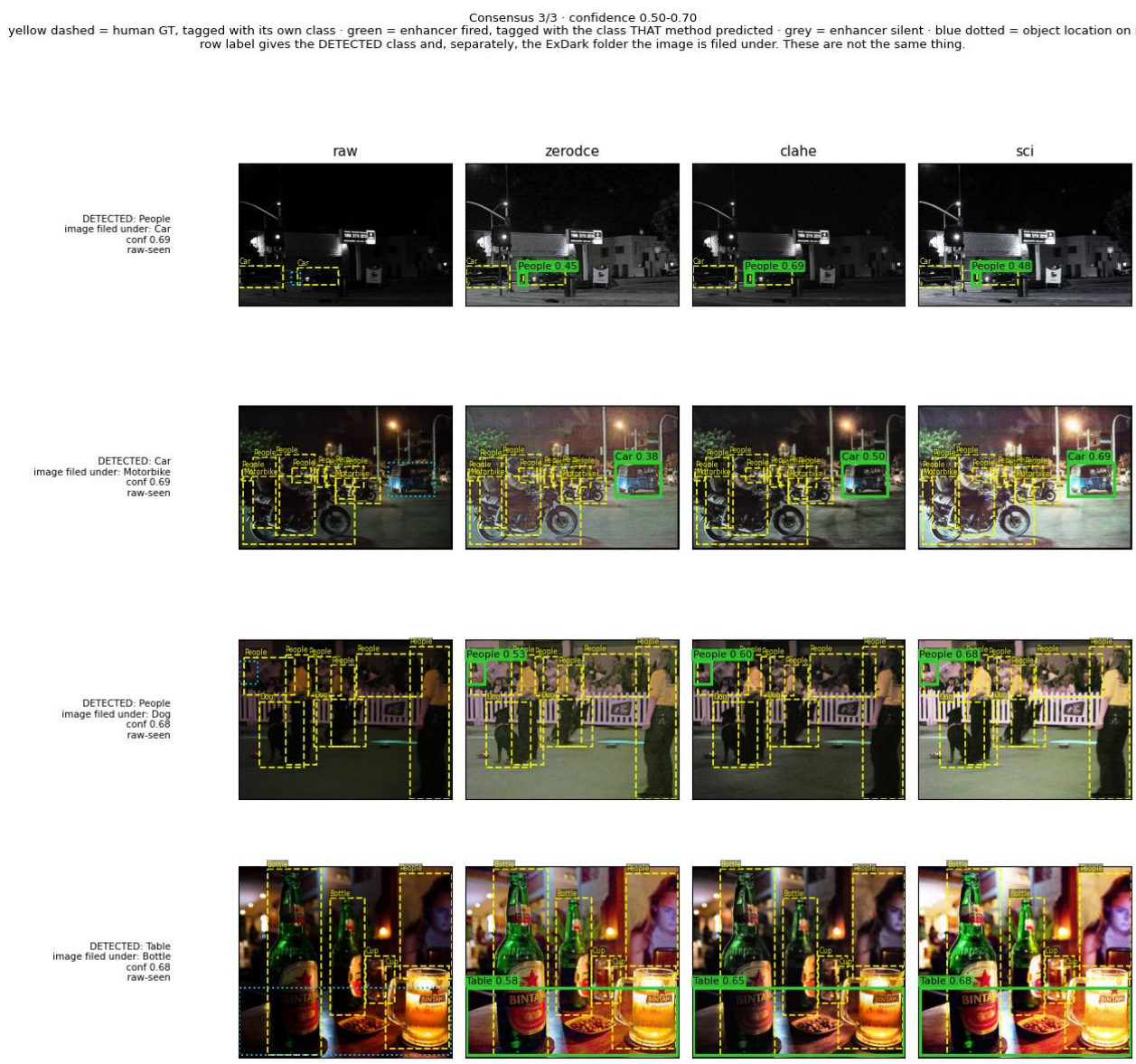


Figure 4.16 Four objects at 3 of 3 consensus and confidence 0.50 to 0.70. 3 out of the 4 predictions (Car, People, Table) represent genuinely correct, individual surfaced predictions.



Figure 4.17 Four objects at 1 of 3 consensus and confidence 0.50 to 0.70. The bottom 2 predictions: Bottle and Table appear to be genuinely correct detections. While the first 2 predictions are too small to be confirmed.

4.4.5 Performance comparison with literature

The model results are compared with three papers in the literature review section that report results on ExDark: Li et al. (2026), Fu et al. (2025), and Liu et al. (2025). The comparison is tabulated in table below, across the key deployment metrics and sorted by mAP@[.5:.95].

Table 4.20 Published ExDark results against this study, ordered by mAP@[.5:.95]

Method	Source	mAP@0.5	mAP@[.5:.95]	Params	GFLOPs	FPS
A	**-**	**0.893**	**0.634**	**31.8 M**	**50.9**	**77.4**
H25	**-**	**0.827**	**0.479**	**31.8 M**	**50.9**	**77.4**
PFNN	Liu et al. (2025)	0.717	0.451	n/a	n/a	n/a
DarkYOLOv8	Fu et al. (2025)	0.698	0.430	2.9 M	6.7	n/a
ZeroDCE + YOLOv5	Liu et al. (2025)	0.705	0.430	n/a	n/a	n/a
DeC-YOLO	Li et al. (2026)	0.831	0.421	10.7 M	9.6	97.6
C@0.55	**-**	**0.758**	**0.420**	**31.8 M**	**50.9**	**77.4**
Zero-DCE + YOLOv8n	Fu et al. (2025)	0.649	0.395	3.7 M	10.8	n/a
DETR	Li et al. (2026)	0.812	0.389	42.2 M	52.6	21.4

Condition A leads on both mAP@0.5 and mAP@[.5:.95], with significantly better performance for the latter metric. It exceeds the best published entry, DeC-YOLO (Li et al., 2026) by 0.062 at mAP@0.5, and by 0.183 at mAP@[.5:.95] versus PFNN (Liu et al.,2025).

H25, trained with human labels on only a quarter of its images, sits second on the strict metric ahead of every published method, while on mAP@0.5 it falls behind DeC-YOLO. C@0.55, which uses no human annotation at all, records 0.420 at mAP@[.5:.95], within 0.001 of DeC-YOLO and above 3 of the external literature entries in the table, while trailing in mAP@0.5 by 0.073 versus DeC-YOLO.

The proposed pipeline recorded an inference speed of 77.4 fps, making it suitable for real-time detection tasks.

CHAPTER 5: DISCUSSION AND LIMITATIONS

5.1 Preliminary phase

The confidence distributions in Figure 4.1 are an early signal that the OWLv2 scores are a weaker filtering signal versus YOLO-World and Grounding DINO scores. Because its correct and incorrect predictions both cluster at the low end of the range, raising the confidence threshold was expected to remove almost all of its predictions, as opposed to only filtering out the incorrect ones. This is the behaviour is quantified Table 4.3, where OWLv2 retains only 265 boxes.

Table 4.1 shows precision increasing with confidence threshold for all models and conditions. This is to be expected, as raising the threshold removes lowest-scoring predictions first, which are most likely to be incorrect. The consequence is that precision on its own cannot separate one teacher from another, since any model can be made to look precise by discarding enough of its own output. This is what motivates the additional constraints applied in Table 4.4 for the final teacher selection, leading to YOLO-World v8 being selected as the teacher model.

5.2 Stage 2: Knowledge distillation and training

Table 4.10 separates two different failure modes. mAP@0.5 accepts a detection when it overlaps the ground truth by half and primarily measures whether the object was found. Meanwhile, mAP@[.5:.95] averages over stricter overlap requirements up from IoU of 0.5 to 0.95, hence it also measures how precisely the box was drawn. Pseudo-label conditions C@0.55 and C@0.60 show an mAP@0.5 gap of approximately 0.14 versus condition A, and a larger mAP@[.5:.95] gap of approximately 0.21 versus this human-label condition. This suggests that the pseudo-label conditions struggle more with localisation versus detection itself.

To further visualise this the tightness variable is introduced as the ratio of mAP@[.5:.95] to mAP@0.5, reported in the final column of Table 4.10. Human-labelled conditions sit near 0.71 and pseudo-label conditions near 0.55. A model that finds objects but draws them poorly will show a lower ratio, which is what the pseudo-label conditions show. The two approaches, human labels versus pseudo-labels, therefore, differ less in what they detect than in how they draw what they detect, a distinction the localisation analysis returns to in Table 4.16.

The human labels introduce a proportionally larger number of bounding boxes, as they include labels that the teacher model may have missed. Table 4.11 shows that at H25, human images are a quarter of the training set but contribute 4,700 of 14,795 boxes, close to a third of the total number of bounding boxes. The human fraction therefore understates how much of the training signal it supplies, which is part of why a quarter of the images recovers more than a quarter of the gap in Table 4.12. Looking at the 0.201 mAP@[.5:.95] performance gap between the zero-shot baseline and the enhanced human-label trained model, a hybrid pseudo-label mix with 10% human labels recovers 14.5% of the gap, and a 25% human labels recover 29.5% of the gap.

Table 4.13 shows that hybrid pseudo labels improve performance differently based on class. The performance of the weakest class, Table, improves the most with hybrid pseudo-labels for mAP@[.5:.95], closing the gap by 64.5% with 25% of human labels. This is to be expected as Table is the worst performing class for the teacher model, recording the lowest per-class F1 in Table 4.7 at 0.124 and the fewest retained boxes at 99.

The same table shows Bicycle, Chair, Bus and Cat recovering only slightly at H10 but improving far more at H25, suggesting a larger number of training labels are required by the trained RF-DETR model to show meaningful detection improvement. The benefit of

human annotation is therefore not evenly distributed across classes. The classes the teacher already covered well need a larger number of training labels before showing significant performance improvement.

5.3 Stage 3: Comparative benchmarking and analysis

When compared against the untrained baseline in Table 4.14, hybrid pseudo-labels prove to be an attractive option, with H25 closing the mAP gap and achieving 0.827 mAP@0.5 versus 0.893 with pure human labels, using only a quarter of human-labelled images. This improvement is less pronounced for mAP@[0.5:0.95], with H25 giving 0.479 mAP@[0.5:0.95] versus human label performance of 0.634, compared to 0.402 from the zero-shot floor. This suggests that pseudo-labels add a larger positive effect to detection, versus localisation.

Figure 4.9 and Table 4.15 show the pseudo-labelled conditions closing the gap between zero-shot and human label condition A, with performance improvement most prominent for Chair up to Boat. Pseudo-labels are most effective for the Bicycle class, where H25 reaches 0.874 against 0.877 for condition A, a difference of 0.003. On the other hand, pure-pseudo labels showed minimal to no gains for classes Bus through Cat, relying on the human labels to improve baseline AP. Table is the weakest class for pure pseudo-label conditions, consistent with the teacher YOLO-World v8's earlier recorded performance in Table 4.7.

A similar behaviour is observed for AP@[0.5:0.95] in the right panel of Figure 4.5, though the gap between the pseudo-label conditions and human label condition is wider, reflecting the localisation challenges faced by both the zero-shot and pseudo-label trained models. The performance gap is the largest for Dog, while pseudo-labels are found to be effective in closing the localisation gap for Table. Taken with Figure 4.9, the classes that

gain most from pseudo-labelling are those the zero-shot model already handled worst, and those that gain least are the ones it already handled well.

Table 4.16 shows the zero-shot RF-DETR drawing a box smaller than the ground truth on 38.4% of its matched detections, with this share decreasing to 32.7% after pseudo-label training and to 5.9% under full human supervision. This convention is likely to be inherited from RF-DETR's COCO pretraining, since the zero-shot model has seen neither an ExDark image nor a pseudo-label. The teacher YOLO-World model loosens these boxes during knowledge distillation, and the addition of human labels further fine-tunes the model's bounding box tightness to match the ExDark ground truth tightness. The median ground-truth to prediction area ratio falls in the same order, from 1.192 to 1.158 and then to 0.994, so the correction is gradual rather than abrupt.

5.4 Image enhancer robustness probe

While Zero-DCE was initially selected as the enhancement method due to its zero-reference nature, Table 4.17 verifies that enhancement choice is not a key factor affecting the object detection performance. All three methods fall below the raw condition on both metrics, and the spread between them, 0.5631 to 0.5677 mAP@[.5:.95], is narrower than the seed-to-seed spread of 0.0050 within the raw condition alone. The choice of enhancer therefore matters less than which random seed a single run happened to use.

On the model level, the lesser than expected benefit of enhancement may be due to a few factors. First, the RF-DETR detector may already be illumination robust, evidenced by the zero-shot RF-DETR mAP@0.5 of 0.732 on raw ExDark images in Table 4.14. The large-scale pre-training for the model may already be sufficient for such object detection tasks. Second, enhancements may only optimise quality for human perception, such as exposure and smoothness, but do not directly improve the features the model relies on for

object detection, such as edges, textures and gradients. Figure 4.12 is consistent with the second explanation, as Zero-DCE and SCI raise brightness while also amplifying sensor noise in precisely the dark regions where additional detail would be most useful.

5.5 Performance comparison with literature

From a parameter count perspective, Table 4.20 shows the proposed models are larger due to their transformer architecture. Compared to lightweight YOLO variants, they carry 31.8 M parameters, roughly 3 times the parameters of DeC-YOLO at 10.7 M and 11 times DarkYOLOv8 at 2.9 M. Their computational cost is 50.9 GFLOPs, between 5 and 8 times more than DeC-YOLO and DarkYOLOv8, respectively. Nonetheless, this architecture is still more lightweight than the classical DETR, that uses 42.2 M parameters and 52.6 GFLOPs, both above RF-DETR-Small. The accuracy in Table 4.20 is therefore obtained at a real cost in model size, and the comparison places this pipeline between the lightweight YOLO derivatives and the classical detectors rather than ahead of either.

5.6 Limitations

The only way to concretely evaluate the true IoU accuracy would be to manually inspect the accuracy of the entire ExDark dataset, perform any ground truth corrections, and re-evaluate the mAP. This is not explored in this research due to the significant resources required for the evaluation. The same constraint applies to the surfaced detections counted in Table 4.18 and Table 4.19, which are identified by the detectors themselves rather than confirmed by eye.

CHAPTER 6: CONCLUSION & FUTURE WORK

This study evaluated the performance of a low-light detector that is trained without human annotation, using offline illumination enhancement and pseudo-labels from a vision-language teacher. The design is decoupled so that enhancement, annotation and training can be varied one at a time. Six conditions are trained on ExDark, alongside an untrained COCO-pretrained model evaluated on the same test split.

Teacher selection is governed by class coverage rather than precision alone. YOLO-World v8 is selected as the teacher model, with its confidence threshold rederived on the full training split as 0.55, against the 0.60 obtained from a 600-image sample.

Full human supervision on raw images performed best at 0.893 mAP@0.5 and 0.634 mAP@[.5:.95], with enhanced human labels marginally behind at 0.883 mAP@0.5 and 0.621 mAP@[.5:.95]. The hybrids followed, at 0.827 mAP@0.5 and 0.479 mAP@[.5:.95] for a 25% of human labels mixed into the pseudo label set, and 0.799 mAP@0.5 and 0.449 mAP@[.5:.95] for 10% of human labels mixed. The models trained with pseudo-labels alone placed lowest among the trained conditions, at 0.758 mAP@0.5 and 0.420 mAP@[.5:.95], for the 0.55 confidence threshold. For the 0.201 mAP@[.5:.95] gap between the pseudo-label-only condition and the enhanced human-label model, a hybrid mix with 10% human labels recovers 14.5% of the gap, and 25% human labels recover 29.5% of the gap. This validates hybrid pseudo-label knowledge distillation as a valid approach for this detection work.

Enhancement produced no measurable benefit in this experiment, where Zero-DCE provided no significant contribution in improving mAP. This is further verified via the enhancement method probe. The enhancement stage can be removed without measured loss, eliminating a preprocessing stage from deployment.

This pipeline outperforms other low-light detection literature on ExDark with its larger transformer architecture. In terms of deployment metrics, the model has 31.8M parameters, 50.9 GFLOPs and performs inference at 77.4 fps.

Future work can be explored in two directions. First, substituting a smaller student would test whether pseudo-labelling leads to larger mAP improvement. The gains from pseudo-labelling versus zero-shot is smaller than expected, as RF-DETR already possesses a heavily pretrained backbone. A smaller student model would also address deployment, as the present model is an order of magnitude larger than the most efficient published, YOLO-based comparator.

Second, an evaluation of the benchmark dataset will be helpful to validate any understated performance, whether in terms of IoU, or new genuine detections revealed from image enhancements. Auditing the surfaced detections, as well as a full validation of the labels in the test image split will fully confirm the validity of the enhancement module, and potential of this low-light image detection pipeline. Evaluation via additional low light datasets will also serve as a supplementary method to validate the performance of the model produced by this study.

CHAPTER 7: APPENDIX

The code notebooks used to produce these results can be found via the Google Drive: https://drive.google.com/drive/folders/1wCni1L4xS_38gXKeiEIRvMf2YVJQsBoy?usp=sharing

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